



Motel Computers Ltd.

THE CYPHER[™] 68000/ Z80

SINGLE BOARD COMPUTER

WITH HIGH RESOLUTION GRAPHICS

TECHNICAL INFORMATION MANUAL

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INTRODUCTION

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CYPHER, from MOTEL COMPUTERS, is a complete 68000,Z80 based single board computer. The 10" X 15" board contains an impressive number of functions which will meet the increased computing requirements of computer professionals and OEMs at a very attractive price.

The main theme of the board is two popular processors (68000 & Z80) supported by 256K ==> 1 MEGABYTE memory and a NEC 7220 graphics display section with its own 128k of display memory . Currently the ONE MEGABYTE CYPHER is the memory size standard for our software since it has the best price performance tradeoff. Important auxiliary functions, namely serial I/O and floppy disk I/O are present, along with other features, that make CYPHER the one board solution to sophisticated 16/8 bit computation.

With CYPHER, one can use 8 and 16 bit technology, together as a team to get your timely work done. For instance using the CP/M operating systems one can edit, using Wordstar, a C source file for the 68000 in the Z80 CP/M 80 mode, and then switch over to CP/M68K to compile and run your 68000 programme. Note, that with the ONE MEGABYTE CYPHER our BIOSs support a 256K operating system for CPMPPLUS, CPM68K A N D a 766 KBYTE RAM DISK !! The 766 K RAM DISK appears identical to both the EIGHT AND SIXTEEN bit operating systems. Thus one can be editing files on the RAM disk with the Z80 and later switch over to the 68000 and compile/assemble files. The operation of CPM68K is greatly enhanced by the 766 K RAM disk and compiling a 500 line CPM68K C programme (three pass compiler, assembly and linkage) takes approximatley a minute. Of course also loading command files is speedy. The RAM disk is large enough that all files necessary for for using DIGITAL RESEARCH's C language for the 68000 can be accomodated with enough space left over for editors, executable code, etc.

If you already have eight bit software and eight bit equipment (editors, data bases, special IO drivers, etc.) the CYPHER can aid you in making an economic transition from using eight bits to sixteen bits while maintaining certain eight bit portions of your programmes/hardware.

To further emphasize CYPHER's capabilities look at its impressive features:

- * DUAL 68000, Z80 PROCESSORS
- * 256K ==> 1 MBYTE PRIMARY MEMORY
- * TWO 28 PIN EPROM SOCKETS & TWO 28 PIN SRAM SOCKETS
- * MEMORY, I/O BUS EXPANSION INTERFACE
- * VIDEO DISPLAY SECTION CAPABLE OF UP TO 1024 X 1024 PIXELS
BASED ON THE NEC 7220 GRAPHICS CHIP
(useful for CAD/CAM and other applications)
- * BATTERY BACKED UP REAL TIME CLOCK (MC146818), PROGRAMABLE INTERRUPT
GENERATION FOR REAL-TIME, MULTITASKING APPLICATIONS
- * TWO Z80 SIO BASED RS232 SERIAL COMMUNICATIONS FOR FLEXIBLE
COMMUNICATIONS OPTIONS, PROGRAMABLE BAUD RATE EACH CHANNEL (8253)
8 & 5 1/4" or 3 1/2 " SINGLE/DOUBLE DENSITY FLOPPY DISK
CONTROLLER (WD2793) UP TO 4 DRIVES, TYPES CAN BE MIXED
- * PARALLEL KEYBOARD INTERFACE (APPLE keyboard compatible)
- * DMA AIDED DATA TRANSFERS (8237)
- * TWO 12 BIT DAC & ADC CHANNELS AND PRECISION VOLTAGE REFERENCE
A TO D DONE WITH SOFTWARE AND COMPARATOR-CONVERSION RATES 5-10KHZ
BASED ON AD7545,AD644,AD581
- * COMPACT 9 1/4 x 14 3/4" MULTILAYER BOARD

CYPHER is flexible in that it is not necessary to have all functions implemented to have a useful working system. For instance one can start with only the Z80 and sufficient memory to have a basic CP/M 80 system. One can add memory and peripherals as required. It is not necessary to use the NEC video section if you already have a terminal. The 68000 can be added at any time and when added takes full advantage of the board's systems resources. CYPHER's unique bus structure allows each processor to access every bit of memory or I/O space in exactly the same way with the exception that the Z80 has a blind 4k window at the top of each dynamic ram bank. A four K window of static RAM appears here to aid bank switching software. The Z80 can select 60k banks.

TECHNICAL OVERVIEW

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Processor switching:

The CYPHER computer is based upon the 16/32 bit 68000 (eight megahertz) processor and the 8 bit Z80A (four megahertz) processor. All system resources are available to both processors but in real time only one processor is active at a time. Virtual co-processing is possible. The board's bus structure is essentially a 16 bit data path supported by an effective 24 bit address bus plus control lines. Switching from one processor to another is accomplished by executing one I/O instruction (Z80) or one memory reference instruction (68000) at the proper port or memory address location. Switching is accomplished through bus arbitration logic.

Memory resources:

There are several types of memory available in the system. There are two 28 pin EPROM sockets which can therefore support a non-volatile R/O storage from 16K Bytes (2764 devices) to 64K Bytes (27256 devices). Primary R/W storage is implemented in Dynamic Ram with total storage ranging from 256K Bytes (4164 devices) to One M Bytes (41256 devices). Additional static ram is available via two 28 pin EPROM/SRAM compatible sockets. A small amount (50 Bytes) of non-volatile R/W storage is available in the real-time clock section. In addition there is 128K Bytes of R/W storage available in the graphics logic section. The graphics memory normally is used for display but could be used in whole or in part to implement an auxillary R/W storage element.

All I/O devices with exception of the 12 bit D/A,A/D ports are 8 bit ports which reside at ODD addresses within the system. The memory and I/O map are dependent upon which processor is active although they are nearly the same for each processor. Each processor has read/write memory extending from address 000000H to address 0FFFFFFH (1 Mega-byte on board). Static ram appears at address FE0000H to address FFFFFFFH for the 68000 and at address XXF000H to address XXFFFFFFH for the Z80. The static ram 'window' for the Z80 is essential for its paging of memory in the implementation of operating systems such as CP/M PLUS . The static ram for the 68000 only appears at address FEF000H to FFFFFFFH. The eprom memory resides from address FE0000H to FFFFFFFH for the 68000 and the Z80. However after a reset eprom memory is enabled at address 0 to enable software startup by the Z80 (the reset time of the 68000 is greater than that of the Z80's).

The Z80 selects it's current 60K page of memory by writing the page address to an eight bit port. This port supplies the upper eight address lines of the 24 bit address bus during Z80 memory cycles and DMA memory cycles.

The non-directly addressable system storage of the VIDEO memory is accessed by

the NEC 7220 through an eight bit I/O port. Transfers can be done by processor control or through a DMA channel. The non-volatile and non-directly addressable memory of the real time clock is accessed also as an I/O port.

The above memory configurations are those which are standard and supported by MOTEL COMPUTERS in current software releases. However system memory can be custom configured to appear at other base addresses. This is possible by re-programming the array logic (PALs) which generate memory selects.

MEMORY MAP at BOOT - UP

System boot up is done under Z80 control. At power up or reset the EPROM memory appears by default at address zero. The Z80 moves code to the static ram window, enables the dynamic ram memory at address zero, and then writes the 68000's 1K byte exception vector table. The Z80 then terminates execution (if a 68000 is present) and waits for the 68000 to come out of reset (the 68000 reset time is greater than the Z80's) and transfer control to it when appropriate.

SYSTEM I/O:

The Z80 addresses system I/O devices as I/O ports. The 68000 addresses I/O devices as memory locations. The top 2 K bytes of the 68000 address space generates I/O control signals which mimic the manner in which the Z80 accesses devices. To the programmer however, the I/O devices are simply memory locations.

Either processor can access and control all I/O devices. A systems programmer may find it convenient to delegate most or all I/O driver routines to the eight bit Z80. This approach eliminates the need to write duplicate driver routines for each processor. However it might be more advantageous in certain circumstances to have the 68000 do all I/O. This could be usefull when implementing multipage Z80 operating systems (ie. CP/M PLUS) since the 68000's linear addressing capability could move data efficiently across 64K byte page boundries for the Z80 (the CPM PLUS bios for the CYPHER does in fact use this technique).

Section 1 CYPHER Functional Blocks

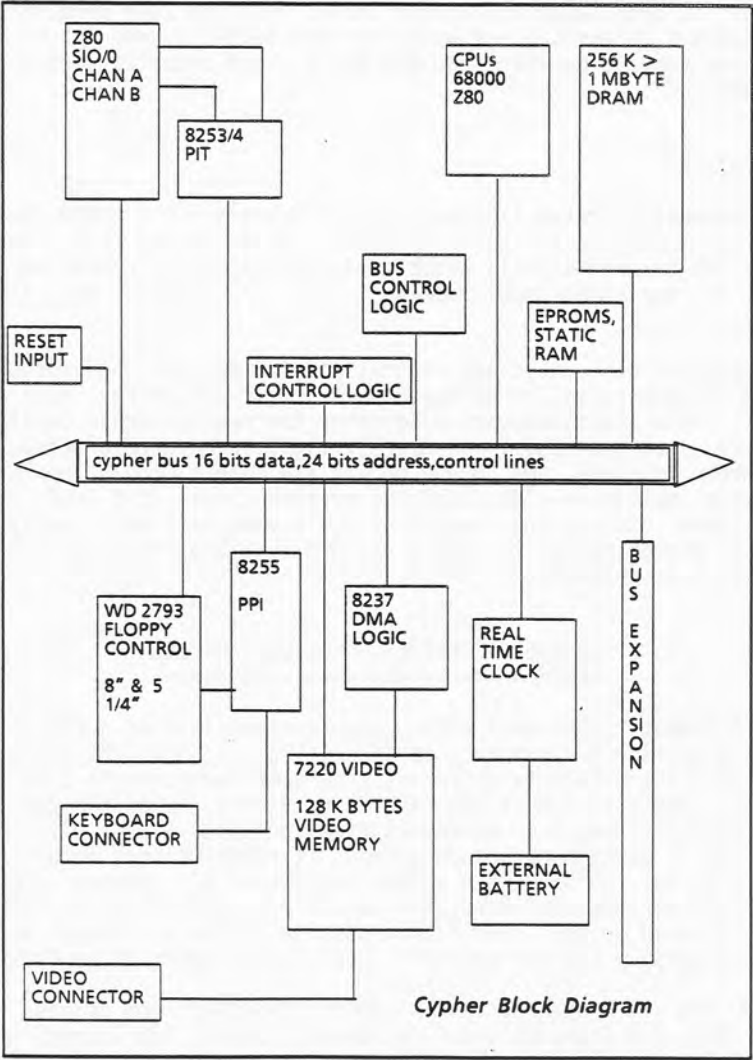
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The CYPHER Computer, like most other computers, consists of a CPU section, memory section, I/O section and graphics section. Fig. 1 shows the block diagram of the system. The CYPHER is different from some computers in that it has two processors. The data bus of the CYPHER is 16 bits. When the Z80 is using the bus it reads(writes) even addresses from the upper 8 data bits ie. D8-D15. When the Z80 is accesssing an odd address it reads(writes) data from the lower 8 data bits ie. D0-D7. This scheme applies to all memory and peripheral devices. All peripherals except the 12 bit D/A and A/D section are eight bits and are accessed on the lower 8 data bits ie. D0-D7. It must be remembered that when accessing I/O devices that they always appear at an ODD address.

The 68000 has a 2 K byte section of memory reserved from address OFFF800H to OFFFFFFH for I/O transactions. For example Serial I/O Channel A Data Port appears at Address OFFF8B1H. The Z80 accesses the same Data Channel at address 0B1H. The DMA channel addressing is slightly more complex and is described in

section 11. When using the bus expansion interface to implement additional peripheral devices it is recommended that they be implemented as memory devices. Call or write for specific details.

* CYPHER BLOCK DIAGRAM *
* *



The CYPHER Board is very complex and only those with some knowledge of practical electronics are advised to attempt its construction. In the world of digital electronics ONE simple error can make the difference between a working board and a non-working one.

One question to be answered before assembling the board is whether one should use sockets or not. Sockets enable the easy removal or replacement of components, such as memory chips, but they are often the most un-reliable part of a computer board. In addition they are an extra cost. Probably a reasonable compromise is to use sockets for the memory chips and more expensive chips such as the NEC 7220 and the MC68000L8. Desoldering chips may damage your board. Remember that the CYPHER is a multilayered board and that desoldering ICs can be more difficult than on a two layered board.

Start of Construction:

The first thing to do before soldering in sockets and components is to visually examine the board for possible shipping damage.

DO NOT SOLDER IN THE OPTIONS SELECTION MALE HEADERS, OTHERWISE THE DEFAULT OPTIONS ETCHED WILL NOT BE ACCESSIBLE! THE CYPHER P.C. BOARD HAS DEFAULT HEADER OPTION LINES. THE DEFAULT OPTIONS APPEARING ON THE COMPONENT SIDE MUST BE CUT IF YOU DECIDE TO SOLDER IN THE HEADERS. THE DEFAULT HEADER OPTIONS HAVE BEEN PLACED ON BOTH SIDES. !!!!!!!!!!!!!!!!!!!!!

Installation of Sockets and ICs:

Use only dual retention dual wiping sockets. Carefully insert the sockets or ICs and be certain that you do not crimp any of the legs. Check that all pins of the IC or socket are poking through to the solder side. Otherwise you may notice half way through soldering a socket a missing pin and be forced to remove the socket and start all over again! Ensure that you have pin one of each socket or IC going into pin one of the corresponding IC position on the board and that you are inserting them onto the component side (indicated by the component identification marks). Pin one of a socket location is indicated by a square pad on both the component and solder sides of the board. The board is four layers with the inner layers being power and ground. There are however some signal traces on the inner layers of the board, so keep this in mind when debugging a board. You may find it more difficult soldering the power and ground pins since they are connected to a large plane of copper metal ie. the power or ground plane. For this reason use extra care in soldering these pins.

Preferably you should solder with a temperature controlled soldering iron that has a reasonably fine tip. Use a fine solder specified for electronic components work. The solder mask should prevent solder splashes from causing signal shorts but as always exercise due care.

When assembling a board it is a good idea to refer to the schematics to get a better idea of the function of the components you are installing. It may not be necessary to strictly adhere to the recommended component values in certain

areas of the board. However before deviating from recommended values be sure you understand why you can make the change.

Installation of Resistor Networks:

Two types of resistor networks are used: SIPs and DIPs. The serial in line package (SIP) resistors are all pullups with pin one of each being connected to Vcc. in the same orientation as the chips. The recommended pull-up value is 2.2K ohms.

Install RN1 (8 pins) at location E1. Pin one is at the top and indicated by the square pad.

Install RN2 (6 pins) at location F1. Spacing around this SIP is minimal so be careful.

Install RN3 (6 pins) at location K1. Due to an error on the Silk-screen this network is erroneously labeled P7. Check for any solder shorts. See Component Layout sheet.

Install RN4 (9 pins) at location E6 just below the 8255 and opposite D1. The label for RN4 is missing. See Component Layout sheet.

Install RN5 (9 pins) at location E10.

Install RN6 (7 pins) at location A13. The location for RN6 is just to the left of IC A14. The label for RN6 is missing. See Component Layout sheet. Install RN7 (6 pins) at location F14 which is just below R22. This network is difficult to solder due to space restrictions. Check for possible solder shorts.

Install RN8 (6 pins) at location L22.

The SIPs are now all installed.

Install the 33 ohm DIP network at socket location L10. This network forms series damping for the address lines going to the Video memory chips. It is imperative that the resistive value be between 22-35 ohms. If desired one could form a DIP network from a component carrier and discrete 1/4 watt 33 ohm resistors.

Installation of Discrete Resistors:

All discrete resistors unless otherwise specified are 1/4 watt.

Install R1 (2.2K)	at location B1
Install R2 (330)	at location F1
Install R3 (220)	at location H1
Install R4 (220)	at location K1
Install R5 (330)	at location L1
Install R6 (33)	at location H2

The positioning of R6 is immediately below IC F2 and RN2. The left hole is just below RN2. The right hole is above the ' 6 ' in 'R6'. The hole is smaller than the left and contains a feed through signal. See Component Layout sheet for details.

Install R7 (150)	at location K2
Install R8 (33)	at location F4
Install R9 (33)	at location K4
Install R10 (220)	at location C6
Install R11 (1K)	at location F6
Install R12 (150)	at location H6
Install R13 (1K)	at location A8
Install R14 (4.7K)	at location C8
Install R15 (2.2K)	at location A9
Install R16 (1.K)	at location A9
R17	not required
Install R18 (1.K)	at location B10
Install R19 (4.7K)	at location C10
Install R20 (2.2K)	at location A13
Install R21 (25K)	at location B14

R21 is missing from the silk-screen. It is between IC @B14 and IC @B15. See Component Layout sheet.

Note that R21 in conjunction with the capacitor C6 at location C15 determines the head loading time for the floppy disk drive interface. The formula for the head load timing is:

$$T_h = 0.32 \times R_t \times C \times (1 + 0.7/R_t)$$

where

T_h = the head load time in MILLISECONDS

R_t is in K ohms

C is in uF

For a head load time of 40 milliseconds approximate values for $R_t = 25K$

$C = 5uF$

A TANTALUM capacitor is recommended for the capacitor. When installing electrolytic capacitors be CERTAIN TO INSTALL THEM WITH THEIR POLARITIES the correct way. Where a capacitor is electrolytic the polarity + - orientation is shown on the Component Layout sheet.

Install R22 (560)	at location E15
Install R23 (2.2K)	at location A15
Install R24 (560)	at location B15

R24 is not labelled but is directly below R23. See Component layout sheet.

Install R25 (1K)	at location B16
Install R26 (330K)	at location H16
Install R27 (22MEG)	at location H16

R27 is not labelled but is directly below R26. See Component Layoutsheet.

Install R28 (2.2K)	at location L16
Install R29 (330)	at location A17
Install R30 (100K)	at location K17
Install R31 (560)	at location L18
Install R32 (330)	at location A18
Install R33 (560)	at location L18
Install R34 (2.2K)	at location A20
Install R35 (20K)	at location L20

(not required)

R35,R36,R38 not labelled. See Component Layout sheet.

Install R36 (1MEG)	at location L20
Install R37 (20K)	at location L20
Install R38 (1MEG)	at location L20
Install R39 (33)	at location A21

Install R40 (33)	at location A22
Install R41 (33)	at location B22
Install R42 (33)	at location C22
Install R43 (2.2K)	at location K22
Install R44 (2.2K)	at location E24
Install R45 (560)	at location L25
Install R46 (560)	at location L25
Install R47 (33)	at location H27
Install R48 (330)	at location H29
Install R49 (2.2K)	at location L29
Install R50 (2.2K)	at location E30
Install R51 (2.2K)	at location H30

→ FOR 2.2K x R49-R51

R52 not required

Install R53 (560)	at location B12	R53 not labelled. See Component
Install R54 (2.2K)	at location L26	" Layout sheet"
Install R55 (2.2K)	at location L28	" "

Diode Installations:

There are five diodes to install and one DUAL COLOURED LED to install.

Install D1 (1N4001)	at location F5
Install D2 (1N4001)	at location C8
Install D3 (1N4001)	at location C10
Install D4 (1N4148)	at location L18

The label for D4 is missing but is just left of R31. See Component Layout sheet.

Install D5 (MBD709)	at location L18
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D5 is a schottky diode. (If MBD709 part not available use 1N4148 instead.)

Install the dual coloured LED (red & green) at location C6. The LED has only two leads. The LED should be installed such that the FLAT part of the plastic package is towards the the connector side of the board. When the board is operational RED indicates Z80 processing and when GREEN 68000 processing.

Capacitor Installation:

Capacitors on the CYPHER board have two functions - to serve as RC timing elements and to minimize power supply noise on the board. The capacitors which are related to the latter function are not numbered and appear as a capacitor symbol on the component layout. The unlabelled capacitors are .1uF bypass capacitors. It is a good idea to install these to reduce the possibility of transient noise on the board especially with regards to the memory section. However the four layer construction of the board lessens the need for these capacitors. Where a value is indicated it is to be a tantalum electrolytic (preferably) with the value shown on the Component Layout sheet. When installing the electrolytics be sure to observe the polarity convention as indicated by the Component Layout sheet.

There is one 60 pF variable capacitor on the board which is used in the WD2793 data separation circuitry . See the section on the floppy disk interface for data separation circuitry adjustment.

Be certain that the capacitors used for timing elements are of the proper value and correctly installed.

Timing capacitors installations:

Install C1 (33pF) at location K29 → 2345 RAM Timing
 Install C2 (.01uF) at location L30
 Install C3 (.01uF) at location H13
 Install C4 (330pF) at location B16
 Install C5 (.01uF) at location A17
 Install C6 (4.7uF) at location C15 Tantalum , observe polarity
 Install C7 (.1uF) at location E5
 Install C8 (60 pF variable capacitor) at location F5

NOTE: Capacitors followed by an asterisk indicates that it is related to Real Time Clock Function. If not using the Real Time Clock do not insert.

Install C9 (50pF) at location L17 *
 Install C10 (1.0uF) at location L17 * Tantalum , observe polarity
 Install C11 (4.7uF) at location L18 * Tantalum , observe polarity
 Install C12 (.1uF) at location L18 *
 Install C13 (50pF) at location H17 *
 Install C14 (50pF) at location H15 *
 Install C15 (.01uF) at location K18 *
 Install C16 (1.0uF) at location C11 Tantalum , observe polarity
 Install C17 (22uF) at location C7 Tantalum , observe polarity

NOTE: Capacitors followed by a double asterisk indicates it is related to A/D, D/A conversion functions. If not using these functions do not insert.

Install C18 (.1uF) at location A11 **
 Install C19 (.1uF) at location A12 **
 Install C20 (.1uF) at location A10 **
 Install C21 (.1uF) at location A12 **
 Install C22 (.1uF) at location B11 **
 Install C23 () at location A8 ** not required
 Install C24 (15pF) at location A10 **
 Install C25 (.1uF) at location A9 **

COMPLETELY ASSEMBLED PARTS LIST

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See Component Layout Sheet for placement.

Integrated circuits

Note: If the video circuitry is operated at or below 20 Mhz the Schottky TTL can be replaced with LS (low power schottky).

Part	Number	Quantity	Part	Quantity
74LS00		3	8 pin SIP 2.2k	1
74S00		1	6 pin SIP 2.2k	4
74LS01		1	9 pin SIP 2.2k	2
74LS02		3	7 pin SIP 2.2k	1
74S02		2	33 ohm 1/4 watt	8
74S04		1	150 ohm 1/4 watt	1
74LS04		4	220 ohm 1/4 watt	5
7406		1	330 ohm 1/4 watt	10
74LS08		4	560 ohm 1/4 watt	3

74LS11	1	1k ohm 1/4 watt	4
74LS14	3	2.2k ohm 1/4 watt	13
74LS20	3	4.7k ohm 1/4 watt	2
74LS22	1	20k ohm 1/4 watt	3
74LS30	1	100 ohm 1/4 watt	2
74S30	2	100k ohm 1/4 watt	1
74LS32	7	330k ohm 1/4 watt	1
74S32	1	1 MEG 1/4 watt	2
		22 MEG 1/4 watt	1
74S37	2	15 pf	1
74LS74	6	33 pf	1
74S74	2	50 pf	3
74LS86	1	330 pf	1
74LS123	1	.01 uf	4
74LS133	1	.1 uf	8
74LS138	2	1 uf	6
74LS148	1	4.7 uf	2
74S163	3	22 uf	1
74LS164	4	47 uf	1
74LS175	2		
74S194	1	60 pf variable cap.	1
74LS240	3	16 MZ crystal	1
74LS245	10	32.768 KZ crystal	1
74LS257	6	25 MZ crystal	1
74S257	1	33 ohm 16 pin DIP (has 8 resistors)	
74LS273	2		
74S299	2		
74LS373	3		
74HC00	1		
MC14069	1		
MC14574	1		
MC146818	1		
AD7545	2		
556	1		
1488	2		
1489	2		
LM311	2		
MC68000L8	1		
Z80A	1		
8203	1		
8237-4	1		
8224	2		
8255A	1		
NEC7220 or Intel equiv.	1		
Z80A SIO/O	1		
8253	1		
WD2793	1		
2764 (provided)	2		
6116	2		
4164-2	48		
PAL1 (provided)	1		
PAL2 (provided)	1		

NOTE: The CYPHER design uses a large number of 74XX257 multiplexors. We have found that certain manufacturers '257s exhibit transient output glitches when switching under certain conditions. We have found however that MOTOROLA and TEXAS INSTRUMENTS devices function properly. We therefore let you draw conclusions about the purchase of these devices. MOTEL COMPUTERS is not implicitly, or otherwise, endorsing the use or nonuse of any manufacturers TTL devices.

Resistor Sips

All resistor SIPS on the board have pin one as common. Pin one is connected to Vcc which then supplies each resistor with five volts that enables it to act as a pullup to the signal which it is connected to. The value of the SIPS can range from 1.1K to 4.7K. If the SIP with the proper number of pins cannot be found it is possible to carefully cut or snap off the extra length of a network having more pins and then install it.

NETWORK NUMBER	RECOMMENDED VALUE	
RN1 (8 pins)	2.2K	F1
RN2 (6 pins)	2.2K	H1
RN3 (6 pins)	2.2K	K1
RN4 (9 pins)	2.2K	F6
RN5 (9 pins)	2.2K	F10
RN6 (7 pins)	2.2K	A13
RN7 (6 pins)	2.2K	F14
RN8 (6 pins)	2.2K	L21

RESISTOR DIP:

The one resistor dip is a 16 pin package of 20-33 ohms. Resistor 1 has connections at pins 1 and 16, resistor 2 at 2 and 15 etc.

CRYSTALS:

There are four crystals on the board.

Crystal one (Y1) is used as a time base for the REAL TIME CLOCK. Its value is 32.768 Khz.

Crystal two (Y2) is used to generate a time base for the 68000 and the Z80. Its value is 16.000 Mhz. Use a SERIES RESONANT FUNDAMENTAL MODE type of crystal.

Crystal three (Y3) is used to generate a time base for the dynamic memory controller chip. Its value can range from 18 to 27 Mhz. We recommend a value near 25 Mhz. A slower crystal will mean a longer dynamic ram access time. Use a SERIES RESONANT FUNDAMENTAL MODE type of crystal. Note that when using frequencies above 20 Mhz it is possible with certain XTALS that they operate in a harmonic of the fundamental. This usually will mean a SLOWER frequency than one expects. If this happens one will have S L O W memory access times and poor performance. Also memory refresh times will be violated (ie memory

integrity cannot be guaranteed). If possible check the clock frequencies of time bases generated by crystals on the board.

Crystal four (Y4) is used to generate the video timing and its frequency is the PIXEL CLOCK RATE. The same precautions apply to this XTAL as the others. See the VIDEO section for more details. A frequency of 20 Mhz is sufficient to generate a 480x220 screen size used to emulate a 80 by 24 line terminal using a 5 x 7 character matrix. For more details on video timing details see the recently released " C " programme which emulates a terminal. This programme has a companion which allows one to dynamically choose frame dimensions and automatically calculates the proper (if mathematically and physically possible) video timing parameters for a standard video monitor (Zenith or similar).

NOTE:
Crystal four (Y4) is used to generate the video circuit timing. With the current oscillator circuit (8224) one can go up to 27 Mhz. The frequency chosen depends upon the characteristics of the monitor one intends to interface to.

With a few modifications it would be possible to bypass the 8224 circuit and employ instead an oscillator package in place of the crystal and the 8224. The maximum useable frequency is approximately 40 Mhz, which is a function of the video controller chip requirements (NEC 7220) and the TTL logic.

REQUIRED HEADER JUMPERING:

Although the bare board has default jumpering. The following is required for a normal working CYPHER. As stated before DO NOT SOLDER IN MALE HEADERS on the jumper areas that have the default shorts otherwise you will never be able to access them again, or change them!!!!

J5 1-2
J6 1-2
J7 2-3
J24 1-2 3-4 5-6 7-8
J26 2-3
J27 2-3

WAKING UP THE BOARD:

There are a few steps involved in getting your board up and running. It is assumed that you will be communicating with the board via a terminal or equivalent device through the RS232 connector P2. Baud rate communications are set for 9600.

Power is supplied through P5 which is an ' IBM ' style power connector. Voltages required are +5,+12,and -12. Carefully study the power connector pinout before applying power. The number of pins of the connector is FEWER than that on the IBM PC. The five volts is the main supply and power consumption can be as high as 6 amps on a fully populated board using schottky TTL in the video section. The +12/-12 is used to power the RS232 interface and the analogue A/D section. If you have used sockets everywhere on the board you probably will like to supply power to the board before inserting any ICs.

You will need a reset mechanism. The first choice is the reset connector while another reset input is possible through the auxillary keyboard connector.

The circuit for the reset switch is shown in the schematics section. If you are not going to use a separate reset switch and rely on the keyboard reset it is important to jumper pins 1 and 2 of P6. Otherwise connect your reset switch to P6 as shown in the diagram. Also if you do not plan to use the 68000 abort switch also connected through P6 it is useful to jumper pins 7 and 8 of P6. On the underside of the board pin 3 of the keyboard connector is connected to pin 8 via a pulldown resistor to prevent erroneous positive resets to the system if no keyboard is attached. If your keyboard does not reset properly with this resistor, remove it. If you are not planning on using the keyboard reset you may want to directly tie pins 3 and eight of the connector together. The point of this matter is that if either one of the unused reset inputs is not properly configured then resets to the system will be disabled and the sytem will not work.

It is assumed that all the discrete elements are installed on the board and all required jumpering, reset switch installation, as described above is complete.

Crystal four (Y4) is used to generate the video time base. Its value might typically range from 15Khz to 27Khz or more. At frequencies above 27Khz it will be necessary to replace the 8224(@H12) with a metal can oscillator.

At this point no ICs are installed nor is any peripheral attached. Supply power via the power connector P5. With a voltmeter check that voltage levels on the board are normal using a proper ground reference ie the ground of your power supply or a ground on the board at the power connector.

You should observe five volts at pin 20 of the socket at location K1. You should observe +12 volts at pin 14 of the socket at location A1. You should observe -12 volts at pin q of the same socket. There should be no problems unless you have improperly installed the electrolytic capacitors. Turn the power off to the board.

The dual coloured LED at location C6 is a useful indicator. Install the 74LS04 IC at location D11. The IC is just to the left of J18 and J17. Now turn the power back on again. The LED should glow green. If it does not you have installed the LED in the wrong direction. If an error has occured correct it.

Install the remaining ICs with the exception of the 68000. If you are only using one bank of dynamic ram be sure to install the chips in the proper banks. Rows D and B are the first bank while rows C and A are the second. Notice that the arrangement may not be what is expected.

The Z80 on power up does several important functions. It moves portions of the monitor into the SRAM from the EPROM and very importantly writes a 1K byte exception vector table for the 68000 starting at location 0.

Now install the ICs and implement the board modifications required as outlined in the modification sheets.

With all ICs required installed (except the 68000) apply power to the board. You should observe a short duration of RED at the LED followed by a permanent GREEN. Use the reset switch to reset the board. Each time reset is asserted you should see the brief RED followed by the GREEN. If you do not an error exists on the board. Carefully retrace your steps if necessary. It is

surprisingly easy to mistake a 74LS02 for a 74LS20. Check for solder splashes between traces and ensure that your modifications are correct.

On power up or reset the Z80 monitor checks if the first byte DRAM in bank 0 is okay before switching to the 68000, if it is not the Z80> prompt will appear (assuming the rest of the board is functional and you have already connected the terminal). If you cannot get the LED to turn green after reset carefully retrace your steps before proceeding further.

If after carefully checking all details and you still cannot get the board to function (even to the green light phase) you can return the board for servicing for a nominal charge . Use whatever tools available to you such as logic analyzers or emulators. Nicolet produces a Z80 emulator for a very low cost and this device generally is sufficient to track down any problems on an ailing CYPHER.

If the LED seems to be indicating a proper system state it is now time to connect your terminal and install the 68000. With the 68000 in place and the terminal connected (to connector P2 at 9600 BAUD) power up the board. You should get the 68000 monitor prompt. Exercise the reset switch. If the prompt appears congratulations, you now have a working 68000 system !

If you get a system prompt but cannot enter data from the keyboard OR characters appear SPONTANEOUSLY on the screen it is because software is polling the parallel keyboard interface. See section 11 on the keyboard interface to remedy the situation.

Assuming the prompt has appeared and keyboard entry appears to be working enter a ' h ' at the keyboard (followed by a return). You should get a listing of the possible monitor commands. Perhaps you would like to explore them now.

If you do not get a proper response from the system (you may get the monitor command list each time you make an entry) it may be that your terminal is not configured to match that of the CYPHER software. Ensure that your terminal is set for 1 START BIT, 8 DATA BITS, NO PARITY, 1 STOP BIT, and 9600 baud.

If you enter an ' s ', for switch you should get the Z80 prompt and notice that the LED has turned RED. RED indicates Z80 processing while GREEN indicates 68000 processing.

At this point you should calibrate the data separation circuitry of the controller and connect your drive(s). See the disk interface section.

Before attempting to boot CP/M 80 you should use the Z80 monitor to READ sectors from your disk drives .

If disk READS appear normal as indicated by the displayed data you should now be able to boot CYPHER/DOS with the configured CYPHER System Disk.

CP/M 68K is booted by reading the CP/M 68K file under CYPHER/DOS. This way the system tracks contain CYPHER/DOS and CP/M-68K is contained in CP68K.COM.

Section 3 Memory/Peripheral Map and Processor Arbitration

=====

The CYPHER's memory and peripheral map may require some familiarization at first, but the structure is quite straight forward considering there are two processors involved.

We will first deal with the memory map and how it may appear to the processors at different points in time.

To the 68000 the memory map IS INVARIANT THROUGHOUT TIME. The memory map to the Z80 IS NOT INVARIANT IN TIME and may take several states.

Below is the memory map of the system as seen by the 68000 at ALL times during its operation.

68000 MEMORY MAP (not to relative scale)

	00FFFFFF HEX	top of 68000 physical address space
PERIPHERAL MEMORY MAP		
	00FFF800 HEX	beginning of I/O
	00FEFFFF HEX	
4K STATIC RAM		FAST STATIC MEMORY
	00FEF000 HEX	
60K EPROM PAGE		68000,Z80 MONITOR UTILITIES PROGRAMMES
	00FE0000 HEX	
approximately 15 Mbytes vacant BERR PRODUCED IF ACCESSED		(BUS EXPANSION REQ'D) WHEN ACCESSED A BUS ERROR IS PRODUCED
	000FFFFFF HEX	ONE MEGABYTE DYNAMIC RAM DECODE RANGE (for use with 256K chips)
	0003FFFF HEX	
256 K BYTES DYNAMIC RAM		
	000003FF HEX	/ initialized by Z80 / on reset or power up
68000 VECTOR TABLE	00000000 HEX	

Z80 MEMORY MAP

	00FFFFFF HEX	top of Z80 paged address space
	00FEFFFF HEX	
4K STATIC RAM WINDOW	00FEF000 HEX	STATIC MEMORY 'TRAVELLING WINDOW'
60K EPROM PAGE		68000,Z80 MONITOR UTILITIES PROGRAMMES
	00FE0000 HEX	
	000FFFFFF HEX	ONE MEGABYTE DYNAMIC RAM DECODE RANGE (for use with 256K chips)
	00040000 HEX	
4K STATIC RAM WINDOW	0003FFFF HEX	
60K DYNAMIC RAM BANK	0003F000 HEX	DYNAMIC RAM BANK 3
4K STATIC RAM WINDOW	00030000 HEX	
60K DYNAMIC RAM BANK	0002FFFF HEX	
	0002F000 HEX	DYNAMIC RAM BANK 2
4K STATIC RAM WINDOW	00020000 HEX	
60K DYNAMIC RAM BANK	0001FFFF HEX	
	0001F000 HEX	DYNAMIC RAM BANK 1
4K STATIC RAM WINDOW	00010000 HEX	
60K DYNAMIC RAM BANK	0000FFFF HEX	
	0000F000 HEX	DYNAMIC RAM BANK 0
	00000000 HEX	

NOTE: If when using the sytem you get a ' BERR ' do not be alarmed. This occurs when you attempt to access non-existant memory above the 1 M Byte address range of the dynamic ram. Accesses to all existant memory should occur without any sytem faults.

Peripheral Decoding and Map:

The Z80 accesses I/O devices through I/O instructions. The 68000 accesses the

same devices through memory reference instructions. Of importance to remember is that I/O devices, with exception of the DACs, are eight bit devices at ODD addresses. For your reference all I/O devices are tabulated below and briefly discussed.

***** PAGE SELECT ***** *Can read bank # back from port 2DH.*

This port is used to set the upper eight bits of the address bus during Z80 processing AND DMA(8237) processing. Either the Z80 or the 68000 can write to this port. When writing to the port from the Z80 it is advisable to be located in memory above 0F000H(Z80 programme counter) in order to prevent a system crash.

PSELECT = 01 (Z80) or FFF801 (68000)

***** NEC7220 *****

The NEC 7220 video controller chip is accessed through an eight bit I/O port. Data transfers can occur under Z80, 68000, or 8237 control. The 7220 control line A0 is connected to BA1 of the CYPHER bus.

7220 = 41H, 43H (Z80) or FFF841, FFF843 (68000)

***** DMA (8237) *****

The 8237 DMA controller is discussed in further detail in the DMA section. Notice below how the Z80 and 68000 address lines are connected to the controller.

8237		Z80/68000
------	--	-----------

A0 <=====	A8
A1 <=====	A1
A2 <=====	A2
A3 <=====	A3

* NOTE: the 8237 must be accessed at an ODD address. A0 is toggled via A8 of the host processor.

When the 8237 controls the bus the upper eight address lines are determined by the state of the PAGE SELECT register mentioned earlier. To aid programming of the 8237 consult the 8237 data sheets and the Z80 BIOS listing.

DMA = 1X(Z80) or FFF81X(68000)

***** REAL TIME CLOCK *****

The real time clock is discussed in detail elsewhere.

RTC = 31, 33(Z80) or FFF831, FFF833(68000)

***** SERIAL I/O Z80SIO *****

The Z80 SIO is discussed elsewhere.

BA1 <=====	C/D* (Z80SIO)
BA2 <=====	B/A* (Z80SIO)

*A data = 81H
A control = 83H
B data = 85H
B control = 87H*

SIO = ~~8X~~(Z80) or FFF8~~8X~~(68000)

→ 81, 83, 85, 87

***** PROGRAMMABLE INTERVAL TIMER *****

The 8253,8254 has two channels dedicated to baud rate generation for the Z80SI0. The third channel is free.

BA1 <=====> A0(8253)
BA2 <=====> A1(8253)

PIT = A9,AB,AD,AF(Z80) or FFF8(A9,AB,AD,AF)(68000)

***** PROGRAMMABLE PERIPHERAL INTERFACE *****

The 8255 is used for disk,keyboard,A/D functions. When programming this device be carefull not to unintentionally alter the states or modes that effect other devices it controls.

BA1 <=====> A0(8255)
BA2 <=====> A1(8255)

PPI = A1,A3,A5,A7(Z80) or FFF8(A1,A3,A5,A7)(68000)

***** DIGITAL TO ANALOGUE CONVERTOR *****

The digital to analogue convertors are twelve bit I/O devices accessed as EVEN memory locations under 68000 control. The least significant bit is connected to BD0 of the CYPHER bus. The most significant bit is connected to BD11 of the bus.

DAC1 = FFF880
DAC2 = FFF888

***** CONFIGURATION *****

The video configuration register is used to divide the frequency of the clock running the video section. Only the lower two bits of this four bit register are used for this purpose. The other two bits are used to control the floppy disk interface.

BD0 <=====> Q0 video timing bit \ divides operating frequency
BD1 <=====> Q1 video timing bit /

BD2 <=====> Q2 SIDE SELECT OUTPUT disk control interface
BD3 <=====> Q3* ENMF* enable mini floppy 2793 disk controller (inverted
Q output)

CONFIG = 99(Z80) or FFF899(68000)

***** ZOOM REGISTER *****

The ZOOM register is used to slow down the rate at which the video shift register clocks out its data. For more details consult the uPD7220 design manual from NEC.

BD0 <=====> Q0* inverted output NOTE: the output bits are inverted
BD1 <=====> Q1* which makes programming zoom
BD2 <=====> Q2* correspond with the needs of
BD3 <=====> Q3* the 7220.

ZOOM = 91(Z80) or FFF891(68000)

***** PROCESSOR SWAP *****

The PROCESSOR SWAP port for the Z80 is OFFH. Any I/O instruction which WRITES to this port address will transfer control over to the 68000. The 68000 will resume processing where it was when it gave up the bus to the Z80. Its internal state is not lost when the Z80 is processing.

PROCESSOR SWAP PORT = OFFH for the Z80 (WRITE operation required)

The PROCESSOR SWAP 'port' for the 68000 is OFFFFFEH. Any memory reference instruction which WRITES data to this port will cause the 68000 to relinquish the bus to the Z80. The Z80's internal state is not lost when the 68000 is in control of the bus.

PROCESSOR SWAP PORT = OFFFFFEH for the 68000 (WRITE operation required)

SOME NOTES ON THE CYPHER BUS CONTROL MECHANISM

The CYPHER uses standard bus control mechanisms to change from processor to processor. The CYPHER board itself has THREE potential bus masters:

- 1) the 68000
- 2) the Z80
- 3) the 8237

When the 68000 is in control the addresses generated on the bus all come from the 68000. When the Z80 or the 8237 is controlling the bus address lines BA16-BA23 are generated from the page port. It is possible for a another device to gain control of the bus through the bus expansion interface by asserting XMASTER* true. There are two unused channels of the 8237 which due to pin limitations of the bus expansion interface are not presented to the interface. These channels are available if the user is willing to make some modifications to the board and interface to them.

Section 4 Interrupt Logic

=====

The interrupt logic on the CYPHER deals principally with the 68000. On the board itself there are three possible sources of interrupts:

- 1) the REAL TIME CLOCK
- 2) the KEYBOARD OR SIO
- 3) the ABORT SWITCH

The interrupt logic is shown in the schematics and allows for seven levels of AUTOVECTOR interrupts(EXCEPTIONS) for the 68000. The REAL TIME CLOCK interrupt(if enabled)is AUTOVECTOR LEVEL ONE. The KEYBOARD interrupt,if enabled is AUTOVECTOR LEVEL TWO. The ABORT SWITCH,if used, is AUTOVECTOR LEVEL THREE. The interrupt levels THREE through SEVEN are presented to the expansion bus for special external devices. In addition a signal called XVPA* (which stands for EXTERNAL VPA*) is presented to the interface which allows an external device the option of using interrupts in the AUTOVECTOR fashion instead of normal exception processing where the external device puts an eight bit vector on the lower half half of the data bus during the interrupt acknowledge cycle. The external device can be implemented to operate either way.

The REAL TIME CLOCK (I1*)interrupt originates from the MC146818 and is normally disabled unless programmed to generate interrupts. The interrupt generated by the real time clock can be caused by several different conditions. The data sheets for the RTC should be consulted.

The KEYBOARD OR SIO (I2*) interrupt has two sources,only one active at a time, which is selected by the JUMPERS J16 and J15(location C13). If J16 is jumpered then the interrupt source is the keyboard , providing the 8255 has been accordingly programmed. If J15 is jumpered then the Z80SIO generates the interrupt. ONLY ONE jumper option should be selected.

The ABORT (I3*) interrupt is generated if an ABORT switch, similar to the RESET is attached. The purpose of the abort switch is primarily for programme debug purposes. Since this interrupt may not be useful for some persons it (I3*) is presented to the expansion bus so that some other device can utilize this interrupt level.

The ONLY Z80 BASED INTERRUPT GENERATED is the MASKABLE INTERRUPT generated by the keyboard.

Section 5 Floppy Disk Interface

=====

The floppy disk interface is implemented with the Western Digital WD2793 LSI controller. Data transfer between the controller chip and the host processors can be done through programmed I/O or through the DMA channel dedicated to the floppy disk interface. Currently BIOS software released for the CYPHER uses the DMA controller. If you are planning on programming the WD2793 you should have its data sheets from Western Digital and if you are using DMA techniques the data sheets from Intel on the 8237. It would be most helpful in addition to examine the MONITOR and BIOS programmes. Also the FORMAT programme written in TURBO Pascal (format source distributed) is a good example of how a high level language can make programming a task normally relegated to assembly language more flexible comprehensive and comprehensible.

The DRQ output of the WD2793 is connected to the DRQ0 input pin of the 8237, in other words channel zero of the 8237 is dedicated to aid transfers to/from the disk controller.

Also, certain I/O functions of the disk control section area are controlled by the 8255 PPI(drive select etc.). See the schematic of the disk control section to aid your programming efforts. Of particular interest may be the motor on bits of the FIVE INCH disk interface. Bit A5 from the PPI controls the motor on signal at pin 16 of P6(called MOTOR ON 1) while bit A4 controls pin 4 of P6. Older versions of the schematics refer to this bit as MOTOR ON 0 whereas its new definition is IN USE. To use these bits jumpers J20 and J21 must be installed. Currently the CPMPLUS compatible BIOS and the format programme support the MOTOR ON bit. Lastly bit 2 of the VIDEO CONFIGURATION PORT is used to control the WD2793 for 5" or 8" drives , while bit 3 is used to select the disk side. The configuration port is a WRITE ONLY port and its condition should be stored in memory so that it can be set up properly when it is being accessed for either Video timing changes or Floppy interface operational changes.

For your reference the following are the addresses of the WD2793, 8255A, VIDEO CONFIGURATION PORT, and the 8237 DMA controller.

***** WD2793 *****		*****	
		Z80	MC68000
STATUS REGISTER	EQU	B9	FFF8B9
COMMAND REGISTER	EQU	B9	FFF8BB
TRACK REGISTER	EQU	BB	FFF8BB
SECTOR REGISTER	EQU	BD	FFF8BD
DATA REGISTER	EQU	BF	FFF8BF
***** 8237 *****		*****	

NOTE: Pin A0 of the 8237 is connected to A8 of the Z80 (and 68000). Register pair BC is used in the Z80 instruction ' OUT (C),A ' to effect a 16 bit I/O address for the system decoding. Because of this fact the addressing of the 8237 is not straightforward. Below are the Z80 addresses of the device. For the 68000 addresses ADD the number FFF800H to those for the Z80.

DMA BASE	EQU	000H+010H
COMMAND REGISTER	EQU	000H+DMA+1001B
MODE REGISTER	EQU	100H+DMA+1011B
REQUEST REGISTER	EQU	100H+DMA+1001B
MASK SET/RESET	EQU	000H+DMA+1011B
MASK WRITE	EQU	100H+DMA+1111B
TEMPORARY REGISTER	EQU	100H+DMA+1101B
STATUS REGISTER	EQU	000H+DMA+1001B
CURRENT ADDRESS	EQU	000H+DMA+0001B (channel Zero)
CURRENT WORD	EQU	100H+DMA+0001B (channel Zero)

For more information see the 8237 data sheets and the DMA section schematics.

***** 8255 *****

Since the 8255 PPI controls certain disk functions we give its addressing as well.

		Z80	68000
PORT A	EQU	A1	FFF8A1
PORT B	EQU	A3	FFF8A3
PORT C	EQU	A5	FFF8A5
CONTROL	EQU	A7	FFF8A7

***** VIDEO CONFIGURATION PORT *****

VIDEO CONFIG	EQU	99	FFF899
--------------	-----	----	--------

DATA SEPARATION ADJUSTMENT:

In order to properly adjust the data separation circuit of the 2793 controller one needs an oscilloscope of at least 20 Mhz bandwidth.

ADJUSTMENT:

Ensure that J6 is open (TEST* pin 22 will be high) and apply a reset to the system (J5 should have pins 1 and 2 always connected). Now connect pins 1 and 2 of J6 which forces TEST* true (active low).

Examine the logic level of pin 37 (DDEN*). If DDEN * is high double all the times mentioned below. The DDEN* is controlled by the 8255 and it should be low after a reset.

Examine the stream of pulses coming from pin 29 (TG43). Adjust the 50K pot, VR2, until the pulse is 1/8 the data rate (250 ns for 8" DOUBLE DENSITY).

500 ns for 5 1/4

With a frequency meter or the oscilloscope observe the frequency on pin 16 (DIRC) and adjust the 60 pf variable capacitor (C8) to give the correct data rate (500 KHZ for 8" DOUBLE DENSITY).

250 KHz for 5 1/4" (4MS)

Observe the pulses on pin 31 (write data). Adjust the 10K variable pot VR3 until they are approximately 250 ns . This adjust may vary from system to system.

(HFM), 500ns for (FM)

Remove the J6 jumper and the disk drive interface should be ready. If you had any problems refer to the Western Digital Manual of the controller chip and ensure that you have properly installed all the discrete components concerned with the data separation circuitry.

5 1/4" - data rate 250 kb/sec

To test pin 16 (DIRC)

- Remove J6 jumper
- Reset
- Insert J6 jumper
- Measure pin 16 & adjust trimmer cap

The two RS232 serial ports of the CYPHER are controlled by a Z80ASIO/0. The baud rate clocks to the SIO are generated from two channels of the 8253(4). The connector P1 allows one to choose the mode of operation of channel A. J23 jumpering allows the use of transmit/receive clocks generated by the 8253 or externally through P1. The jumpering of J24 allows channel A to be either a slave or master device. See the serial interface schematics.

The software released for the CYPHER currently uses channel B(P 2) for terminal communications. The settings for communication with channel B are:

- 1) baud rate = 9600
- 2) no parity
- 3) start bits = 1
- 4) stop bits = 1
- 5) data bits = 8

If J15 is jumpered then interrupts generated by the SIO will be sent to AUTOVECTOR interrupt 2 (I2*) of the 68000. This interrupt also goes to the maskable interrupt of the Z80. When programming the Z80SIO be certain to reference the proper data manuals. Zilog has an excellent reference in a manual called 'MICROPROCESSOR APPLICATIONS REFERENCE BOOK VOLUME 1' dated July 1981. The Z80 has a special RETURN FROM INTERRUPT instruction RETI which informs the Z80SIO of the completion of interrupt handling. When using the 68000 and the Z80SIO with interrupts it will be necessary to write a special return from interrupt command (038 hex) into the WRO of channel A to replace the RETI of the Z80.

PIT (AFH)

Bit 0 = Binary / BCD
 0 = binary 16 bit counter
 * 1 = BCD counter

1,2,3 = Mode (use mode 3)

5,4 = Read-load, 00 = counter latch operation
 01 = R/L least sig. byte only
 10 = R/L MSB only
 11 = R/L LSB then MSB.

7,6 = Counter Selection, 00 = Counter 0
 01 = Counter 1
 10 = Counter 2
 11 = Counter 3

Ports

Timer 0 = A9H (FFFB9)
 1 = ABH (FFFB B)
 2 = ADH (FFFB D)
 Control Reg. = AFH (FFFB F)

Programming

- 1) Output control word to PIT (AFH)
- 2) Send LSB to timer
- 3) Send MSB to timer

Baud Rate

Baudrate BCD value

$$= 2^6 / \text{divisor} / \text{baudrate}$$

$$\rightarrow \text{divisor} = 1, 16, 32, 64$$

$$\rightarrow \text{say decimal answer} = 104.$$

then send LSB (04), then

MSB (10) → don't convert to hex!

if divisor = 16

		LSB	MSB
75	= 1666	= 66	16
300	= 417	= 17	04
600	= 208	= 08	02
1.2k	= 104	= 04	01
2.4k	= 52	= 52	00
4.8k	= 26	= 26	00
9.6k	= 13	= 13	00
19.2k	= 6.5	= 6	00 (= 20.8k)
31.25k	= 4	= 04	00

* SyncA tied to 8253 timer #3, so can be used as a timer interrupt when SIO enabled for external status interrupts.

Section 7 Direct Memory Access (DMA)

=====

The CYPHER has DMA capability through the use of an 8237 type DMA controller. The uses of the controller are:

- 1) move data to/from disk controller
- 2) move data to/from video controller
- 3) move data from memory to memory
- 4) possible special user applications

The 8237 type was chosen for availability, small package size, simple video interface and reasonable performance. The 8237 interface is unusual in that it is a BYTE wide device interfaced to a WORD wide bus structure. Byte swapping occurs when the 8237 operates in a similar fashion to the Z80 operation on the 16 bit bus.

The 8237 interface has a special ' DTACK* like ' timer circuit to ensure that data from the peripheral is not transferred until it is stable. This circuit is shown in part in the SCHEMATICS: DTACK* GENERATION. Depending on the speed of the peripheral the jumper timing can be adjusted. This jumper is normally default jumpered.

The 8237 requires a large number of address lines (four) in order to address all of its registers. That leaves three address lines for Z80 I/O decoding which allows for only eight peripheral devices. Since the CYPHER has more than eight peripheral devices an extra address line was needed. Address line A8 from the Z80 & 68000 processors is used to control the A0 input of the 8237. This is somewhat non-standard but overcomes the address problem. The only penalty for this arrangement is that the Z80 must use the I/O instruction type OUT (C),R in order to communicate with the 8237. This instruction type is a 16 bit I/O type for the Z80 since register C supplies A0 to A7 while register B supplies A8 to A15.

Section 8 Bus Expansion Interface

=====

The design philosophy of the CYPHER was to have a powerful COMPLETE single 16/8 BIT computer board thereby reducing complexity and hence cost. However it is possible to add more memory and peripherals with the expansion bus connector. At the 60 pin expansion interface (which is located near the two EPROM sockets) the buffered ADDRESS, DATA, and CONTROL singles are available.

The signals, their description and pin numbers are given below. Although it was previously mentioned that the signals are buffered it is HIGHLY recommended that any bus expansion board be FULLY buffered at its interface to the expansion bus. To determine which section of memory and I/O space is available for offboard use see SECTION 3 which details the Memory and I/O space.

Note: All signals followed by an asterisk are ACTIVE LOW.

EXPANSION BUS INTERFACE DEFINITION		
Signal	Description	Pin Number
PRES*	processor reset (system reset)	27
A0*	Z80 address 0 (active low)	35
BA1	address 1	51
BA2	address 2	49
BA3	address 3	47
BA4	address 4	45
BA5	address 5	43
BA6	address 6	41
BA7	address 7	39
BA8	address 8	37
BA9	address 9	7
BA10	address 10	15
BA11	address 11	14
BA12	address 12	13
BA13	address 13	12
BA14	address 14	9
BA15	address 15	10
BA16	address 16	1
BA17	address 17	2
BA18	address 18	6
BA19	address 19	4
BA20	address 20	3
BA21	address 21	8
BA22	address 22	5
BA23	address 23	11
BD0	data 0	22
BD1	data 1	24
BD2	data 2	26
BD3	data 3	25
BD4	data 4	23
BD5	data 5	21
BD6	data 6	20
BD7	data 7	18
BD8	data 8	54
BD9	data 9	56
BD10	data 10	58
BD11	data 11	57
BD12	data 12	55
BD13	data 13	53

BD14	data 14	52
BD15	data 15	50
E*	68000 E clock	17 - 68000 Signal
DTACK*	68000 DTACK	36
XMASTER*	External Bus Master Request	34
MEMR*	Memory Read Control	28
MEMW*	Memory Write Control	30
IOW*	I/O Write Control	38
IOR*	I/O Read Control	32
I7*	68000 Interrupt Level 7	46 - Highest priority.
I6*	68000 Interrupt Level 6	44
I5*	68000 Interrupt Level 5	42
I4*	68000 Interrupt Level 4	40
I3*	68000 Interrupt Level 3	48
XVPA*	68000 External VPA	60 - 68000 Signal
VMA*	68000 VMA	17 - 68000 Signal
INTA	System Interrupt Acknowledge	16
Vcc	+ 5 Volt Logic Power	59
Vss	System Power Return	29,31

Note: An omission in the interface are the signals UDS* and LDS*. These signals are only necessary if adding more R/W memory. If you are going to add more R/W memory then two lines of the interface (ie. one of the interrupt lines) can be disabled and jumpered with UDS* and LDS*. See the complete schematics.

The real time clock on the CYPHER is the MC146818 which is a CMOS device that has 64 bytes of read/write memory. The device is supported by a battery backup circuit so that time is kept even when the computer is turned off.

The MC146818 was designed to work with multiplexed address and data buses. However the CYPHER bus is non-multiplexed. The circuitry on the CYPHER overcomes this problem by allowing one to write an address to the clock and then read or write from that addressed location. Therefore to communicate with the clock one must transfer two bytes of information for each access. In order to programme the clock refer to the data sheets. The logic interface truth table to the clock is outlined in the following table.

STATE	CYPHER BUS				MC146818				CLOCK ADDRESS
	IOWD*	IORD*	RTC*	BA1	+	AS	DS	R/W*	
1	X	X	1	X	+	X	X	X	
2	0	1	0	0	+	0	1	1	031H or FFF831H
3	0	1	0	1	+	1	1	0	033H or FFF833H
4	1	0	0	0	+	1	1	1	031H or FFF831H
5	1	0	0	1	+	1	0	1	033H or FFF833H

Note that there are five described states in the above truth table. Of these states only 2, 3, and 5 are of significance. State 2 is used to write an address into the clock. The address is latched on the falling edge of the AS signal. State 3 is used to write data into the clock, assuming an appropriate address has already been setup from state 2. State 5 is used to read data from the clock assuming that an address has already been setup from state 2.

***** Battery Backup Circuit *****

Most of the chips in the REAL TIME CLOCK (RTC) circuit are powered by a special voltage source called BBV. These chips include :

- 1) MC146818 RTC
- 2) MC14069 inverter used as a 32.768 Khz oscillator
- 3) 74HC00 used to disable or enable CE* to the RTC
- 4) MC14574 comparator used decide when to enable the RTC

Refer to the schematic on the REAL TIME CLOCK LOGIC.

The battery is connected via P9 and consists of two header pins. It is up to the user to decide what type of battery to connect. Pin 1 is + in and pin 2 is - or ground. The voltage of the battery can range from 3.5 to 5.0 volts, but it is left to the user to determine the optimum for their particular conditions. The MBD709 diode is recommended to minimize the voltage drop from the main power supply to the Vdd input of the real time clock to prevent the possibility of the logic level inputs to the RTC being higher than the voltage source (Vdd).

The MC14574 comparator is used to enable the CE* input of the RTC. Normally the negative input is LOWER than the reference applied to the positive input which is a function of the main power supply. When this condition is met the output is positive and enables the RTC. When the main voltage drops the

positive input will fall below the negative input (maintained by Vbb) and the RTC will be disabled. Disabling the RTC during power down conditions prevents spurious reads and writes to the chip during the power down and up phases. It also minimizes power consumption by the RTC since when the chip is disabled the inputs are electrically isolated from the rest of the system.

When the user installs their particular battery it will be necessary to adjust the one MEG pot which determines when the RTC will be operative. This procedure can be accomplished by means of a volt meter or oscilloscope used to monitor the state of pin 13 (CE*) of the MC146818 RTC. When power is applied to the system (5 +/- 0.25 volts) CE* should be low. With the battery connected (P9 pin 1 = +, pin 2 = gnd) and the main power turned off pin 13 should be high.

A simple way to calibrate is to observe the state of pin 13 (RTC) while adjusting the 1 MEG pot with the POWER ON. You should get CE* to toggle while adjusting the POT. A reasonable approach is to turn the pot just a small distance past the point where CE* toggles high. Of course the user may choose more exacting methods by measuring the voltages going to the comparator and set these to suit that persons voltage criteria.

The ACCURACY of time keeping can be adjusted by varying the value of C14 and C13. Resistor R26 can also be adjusted. Time keeping accuracy is a function of the crystal, the components mentioned above and the battery backup voltage.

Note that the CYPHER CPM PLUS BIOS fully supports TIME fuctions via use of the RTC. Source listing is available from MOTEL COMPUTERS LTD.

```

ENBLE:  MVI      A,OBH
        OUT      REGADD
        MVI      A,OFH
        OUT      REGDAT
        RST      7
;
;
;
;
TIME:   DB      0,35,23,3,18,9,84
        END

```

Note: If you try to programme the real time clock using DDT you may erroneously conclude that the interface does not work. This is because the RTC interface changes its address with ANY IO cycle. Thus only RTC IO cycles should be allowed between an address set up and data read/write. Using the line trace mode of DDT will violate this condition since an IO cycle to the terminal will occur between the address setup to the RTC and the data read/write cycle. When this happens a wrong address will be strobed into the RTC. Avoid doing this(easily accomplished by a G command).

The CYPHER computer has the capability to generate two channels of ANALOGUE signals. It also can DIGITIZE two incoming analogue signals by comparing the incoming signals to those generated. The DIGITIZATION is done by software successive approximation while the digital to analogue conversion is done by two AD7545 DACs. There is also a 10 volt precision reference source useful in both D/A and A/D applications

The digital to analogue (D/A) and analogue to digital (A/D) section of the computer was designed to work exclusively with the 68000. This was done since the process involves twelve bits. Although it is possible to programme the DACs from the Z80 the results will be unpredictable since four bits will be in an undefined state.

The analogue devices employed in the circuit are:

- 1) two 12 bit multiplying DACS (ANALOG DEVICES AD7545)
(300 ns delay from digital input change to 90 % analogue output)
- 2) Dual High Speed Bifet Op Amp (ANALOG DEVICES AD644)
(13V/us slew rate)
- 3) High Precision 10 Volt IC voltage reference (ANALOG DEVICES AD581)

There are two jumpers and one connector associated with this section of the board.

J26 allows one to input a 10 volt reference signal to DAC # 2 (B8) or to input a possibly varying input signal. In the former case one would be using the DAC to produce a known voltage level while in the latter case one would be using the DAC to MULTIPLY the incoming signal (volume control).

J27 does the same function for DAC # 1 (B10).

The connector P7 allows:

incoming signals for multiplication (3,4 1,2)
outgoing generated analogue signals (9,10 5,6)
incoming signals for digitization (11,12 7,8)

The EVEN pins on P7 are all ground (common).

The address's of the DACs are:

DAC # 1 (B10) = OFFF880H
DAC # 2 (B8) = OFFF888H

In order to DIGITIZE the unknown signal is presented to one input of a comparator while a known signal generated from a DAC is presented to the other input. The output of the comparator is sent to a bit of the 8255 parallel port. Since there are two channels there are two comparators and two bits of the 8255 utilized. Channel 1 goes to bit C4 of the 8255. Channel 2 goes to bit C7. When doing A/D conversions the 8255 must be programmed to have bits C4 and

C7 in an input mode.

***** C A U T I O N *****

When programming the 8255 to have bits C4 and C7 in output mode do not alter the mode of PORT A or disk control will be made inoperative. In two sample programmes that follow shortly a recommended control word for the 8255 will be presented.

CONVERSION TECHNIQUE

Refer to the DAC/AD INTERFACE and FLOPPY DISK INTERFACE schematic to aid understanding of what is to follow. The jumpers J26 and J27 would have pins 2&3 connected to provide the precision reference voltage to the DACs. The analogue signal to be ascribed a digital value is input to pin 3 (- input) of the comparator via the connector P7. An assembly language programme then outputs digital values to the DAC and inspects the output of the comparator to see if it is toggled.

The most efficient method of testing the bits is to start with the most significant and proceed to lesser significant bits. Since the DAC is twelve bits there are twelve bits to be tested. Assuming a test time of approximately 10 us per bit the conversion rate would be $1/12 \times 10 \times 10^{-6}$ or approximately 8 KHz. If one were to sacrifice the bit resolution then the rate would be higher.

It should be pointed out that the board does not have a sample and hold circuitry. If the signal is changing at a rate near the digitization rate then results would be erroneous. This could be remedied with an external sample and hold. If you are measuring signals that your equipment produces then this would be an easy addition to your circuit.

A simple book which explains conversion techniques is MICROCOMPUTER-ANALOG CONVERTER Software & Interfacing (Blacksburg continuing education series/ publisher Howard W. Sams & Co. Inc. 4300 West 62nd St. Indianapolis, Indiana 46268).

SOFTWARE A/D CONVERSION ROUTINE

Accuracy = 12 bits or one part in 4096
Conversion rate = 5.5 KHz or 15 us per bit

The address of DAC # 1 = 0fff880H

The address of DAC # 2 = 0fff888H

```
PPI_CTRL      EQU      $FFF8A7
PPI_MODE      EQU      $8B
PORT_C        EQU      $FFF8A5
DAC1          EQU      $FFF880
DAC2          EQU      $FFF888
BITS          EQU      12
```

NOTE : The output of the comparator is read in fro, a bit of the
8255 Port C.

BIT 4 OF PORT C = INPUT BIT FOR CHANNEL TWO
BIT 7 OF PORT C = INPUT BIT FOR CHANNEL ONE (must be jumped)

register d0 ==> 12 bit result
register d1 ==> test bit
register d2 ==> number of bits to test
register d3 ==> comparator test

PORT_C ==> COMPARATOR PORT

DAC	LEA	PPI_CTRL,A0	* SET UP 8255 PROPERLY
	MOVE.B	#\$8B,D1	* PUT C PORT IN INPUT MODE
	MOVE.B	D1,(A0)	
	LEA	DAC2,A0	
	LEA	PORT_C,A1	* PORT C BYTE ADDRESS
	CLR.L	D0	* CLEAR THE DAC ,OUTPUT TO ZERO
	CLR.L	D1	
CLEAR	MOVE.W	D0,(A0)	
INIT	MOVE.W	#\$800,D1	* TEST PATTERN
	MOVE.W	#BITS,D2	* BIT COUNT IN D2
LOOP	OR.W	D1,D0	* GET BIT TO TEST
	MOVE.W	D0,(A0)	* SEND CURRENT VALUE TO DAC
	MOVE.B	(A1),D3	* GET COMPARATOR DATA
	BTST	#4,D3	* CHECK COMPARATOR STATUS
	BEQ	OK	* IF NOT TOO LARGE TEST NEXT BIT
TOOBIG	NOT.W	D1	* MASK OUT BIT JUST TRIED
	AND.W	D1,D0	* BIT UNDER TEST NOW CLEARED
	NOT.W	D1	* RESTORE D1
OK	ROR	#1,D1	* SHIFT DOWN TO NEXT TEST BIT
	SUBQ	#1,D2	* ALL BITS TESTED YET ?

BNE LOOP

★

TRAP #5

★ CONVERSION FINISHED, TRAP TO
MONITOR AND DISPLAY RESULT

★

★

END DAC

Section 11 Keyboard Interface

=====

The CYPHER has a parallel data input socket (16 pin) location E8 which was designed to be compatible with Apple II type keyboard devices. The primary intention of the socket is as a keyboard input port when using the 7220 display graphic section to emulate a terminal. Other uses are possible however, especially if one is using a separate terminal communicating via one of the serial data ports.

Pin Definitions :

pin 1	= +5	pin 16	= NC
pin 2	= STROBE	pin 15	= -12 Volts
pin 3	= RESET	pin 14	= NC
pin 4	= DATA 7	pin 13	= DATA 1
pin 5	= DATA 5	pin 12	= DATA 0
pin 6	= DATA 4	pin 11	= DATA 3
pin 7	= DATA 6	pin 10	= DATA 2
pin 8	= GND	pin 9	= NC

The keyboard data interface is managed by an 8255 programmable peripheral interface chip. The base address of the PPI is A1H for the Z80 and FFF8A1H for 68000. For the Z80 all peripherals must be addressed at odd addresses since they are eight bit devices which are connected to the lower data byte of the ' 68000 type ' bus. To summarize the 8255's addressing scheme:

** FUNCTION **

PORT A	= 0A1H (Z80) or FFF8A1H (68000)	Floppy Disk I/O Control
PORT B	= 0A3H (Z80) or FFF8A3H (68000)	Keyboard I/O
PORT C	= 0A5H (Z80) or FFF8A5H (68000)	Keyboard & Floppy Ctrl
CONTROL	= 0A7H (Z80) or FFF8A7H (68000)	Configuration

The 8255 is used to control two peripherals, the keyboard and control signals related to the floppy disk drive interface. If the user is to use the keyboard port in a non-standard fashion then care must be exercised in not disrupting floppy-disk I/O functions through an inappropriate choice of set-up modes for the 8255.

The 8255 nominally is set up for MODE 1 operation. Port A is an output port and port B is an input port. Port C has a slightly more complicated arrangement. Bit PC2 is used as an active LOW strobe input from the keyboard. When PC2 (strobe*) goes low data is loaded into the B port from the keyboard. If INTE B (controlled by bit set/reset of PC2) is enabled then bit PC0 (INTRb) will go high which if jumpered at J16 will generate an interrupt to the 68000. The INTRb signal goes inactive as soon as data is read from the 8255. It is possible for software to poll the keyboard port for a keyboard entry. This is done by simply reading port C and inspecting bit PC1 (IBFb). This bit is the INPUT BUFFER FULL F/F and will be set after a strobe* on PC2. It is reset after the 8255 is read.

The other bits on Port C are used as follows:

PC1 ==> NC
PC3 ==> NC
PC4 ==> input from channel A comparator (DAC/AD)
PC5 ==> input from 8 " floppy connector pin 6 FAULT
PC6 ==> input from 8 " floppy connector pin 10 TWO SIDED
PC7 ==> input from channel B comparator (DAC/AD)

NOTE: PC7 can be jumpered if necessary (J22 1-2) for FAULT RESET to pin 4 of the 8" FLOPPY connector. To be utilized it would be necessary to have Port C (upper bits) in a nominal input mode. A fault condition would be detected at PC5. If a fault occurred then it would be necessary to put Port C (upper bits) in an output state to temporarily toggle PC7 to clear the fault condition.

NOTE: Some versions of the monitor check the status of the " B " port of the 8255 for parallel keyboard input at POWER UP or RESET. If there is no parallel keyboard attached BUT the state of the strobe line (JUMPER 18) is LOW the software will erroneously conclude that there is a parallel keyboard attached. This will manifest itself as a continuous string of characters being printed on the screen OR no entry response from a terminal (assuming you are using one). Fix the problem by jumpering 2-3 of J18 and applying a pulldown resistor (220 ohms) from pin 2 of the keyboard port to ground.

Some Z80 sample code is shown below for using the keyboard port.

```
.Z80
PPI_CONTROL EQU 0A7H
PPI_MODE EQU 086H ; port B is input while in mode 1 with lower four
                  ; bits of port C used as control signals. port A is
                  ; output while in Mode 0 with upper four bits of port
                  ; C used in input mode.
inte_mode equ 00000101b ; set bit PC2 to enable interrupts
port_a equ 0A1H
port_b equ 0A3H
port_c equ 0A5H
;
PPI: ld a,PPI_MODE
      out (PPI_CONTROL),a
key_entry: in a,(port_c); check for and get valid keyboard entry
           bit 1,a ; check INPUT BUFFER FULL BIT
           jp z,key_entry
           in a,(port_b); data received so get it
           ret ; and return
;
; enable interrupt to 68000 on keyboard entry
key_int: ld a,inte_mode
          out (ppi_control),a
          ret
;
; disable interrupt to 68000 on keyboard entry
no_key_int: ld a,00000100b
            out (ppi_control),a
```

The CYPHER uses the NEC 7220 (second sourced from Intel) which is a powerful graphics device in its price category. It is an absolute must to have both the data sheets for this device and the uPD7220/GDC DESIGN MANUAL if one is going to programme it. Intel also supplies literature on the device which is very similar to that of NEC's.

The CYPHER implementation of the graphics is as one image plane with the capability of up to 1024X1024 image pixels. The video interface consists of separate HORIZONTAL SYNC, VERTICAL SYNC, and BLANKING signals all of which are active LOW true. The VIDEO output is active HIGH true. The LIGHT PEN input is active LOW.

In the Appendix ' Schematics by Function ' is given a simple circuit which will combine the three separate video signals into a COMPOSITE signal suitable for most low cost computer type monitors.

The clock generator provided for video timing is an 8224 plus a crystal. The maximum rated clock frequency for the 8224 is 27 MegHz. The user may change the frequency of the crystal to suit their needs. If frequencies higher than 27 Meghz are required it is possible with some simple modifications to replace the 8224 with a metal can crystal oscillator package of the required frequency. Pin 12 of the 8224 is the clock output, pins 14&15 are the crystal inputs, while pin 8 is ground and pin 16 is plus five volts.

The CYPHER comes configured with a 20Mhz video crystal. This is adequate for most applications. The EPROM ADM-3A/TEK4010 emulator also has additional CYPHER native commands, including one which can AUTOMATICALLY configure/-reconfigure your terminal (see the next section). There are two registers associated with the video circuit which control the ZOOM magnification and the clock rate of the system. The first is called the ZOOM register (four bits), the second the CONFIG register (two bits). The automatic configure command sets the CONFIG register according to requirements. It also determines whether or not to use the interlace mode of the 7220.

The CONFIG register is a four bit register latched from data bits BDO-BD3. Only the two lower bits of this register are used in the video portion of the circuit. The effect of this register is to alter the effective clock rate of the video circuit. Normally all ones (03 hex) would be written to this register which supplies an undivided time base to the video circuit (full speed). A value of two written to the register slows the time base by a factor of two and so forth down to a zero value which slows the circuit by four times. The other two bits are used to control the disk drive interface. Refer to the NEC manual with details on ZOOM timing (note that the EPROM resident terminal emulation code uses the software zoom function and not the hardware method). The ZOOM register on the CYPHER is implemented in a functionally equivalent manner that is described in the manual except that the four ZOOM bits (latched from BDO-BD3) are inverted. If one writes a ZERO to the register then the ZOOM timing is for 1X ZOOM. A ONE written to the register supplies 2X ZOOM, etc. all the way to 15 which supplies a 16X ZOOM factor.

NOTE: The following information has generally been superceded by the EPROM RESIDENT ADM-3A and TEKTRONIX 4010 terminal emulation package.

There are two programmes supplied on the distribution disk pertaining to the video section. One called GCYPHER.COM intializes the video circuit. However it is unlikely that these parameters would match a users CRT. There are eight parameters in this programme that are relatively easy to change by the user. Refering to the GDC manual section 4.24.3 INITIALIZATION, the parameters 'P1, P1, P2, P3, P4, P5, P6, P7, P8' are given and these appear at a table starting at location 01C7 hex in the GCYPHER.COM file. By changing these values to suit your requirement it would be possible to set your monitor up properly.

Although one can carry out calculations to arrive at parameters to suit a particular CRT and desired screen format the final ' TWEAKING ' of the parameters may be a matter of trial and error.

Another programme supplied can be used to test the operation of the memory in the video circuit. If the GDC memory passes this test then the video circuit should be able to function properly.

See the next section on FIRMWARE for a description of the EPROM RESIDENT graphics drivers.

Our Firmware is constantly being enhanced. The following is a summary of our latest firmware. Refer to CYPHER diskette for software notes.

68000 Monitor: THE FOLLOWING IS DISPLAYED BY THE 68K MONITOR IF AN INVALID COMMAND IS ENTERED.

COMMAND SET:	syntax
D(UMP)	D<START>,<END>
M(EMORY)	M<ADDRESS>
F(ILL)	F<START>,<END>,<BYTE>
C(OPY)	C<SOURCE_START>,<SOURCE_END>,<DEST_START>
V(ERIFY)	V<SOURCE_START>,<SOURCE_END>,<DEST_START>
G(OTO)	G<ADDRESS>
S(WITCH)	S (switch to Z80 monitor)
A(CCEPT)	A<OPTIONAL OFFSET TO LOAD ADDRESS>
R(EGISTER)	R<REG IDENTIFIER> <VALUE> (all reg. display)

(Note: CPM-80 OR CYPHER-DOS is used to boot up CPM-68k)

Z80 Monitor: THE FOLLOWING MENU IS DISPLAYED BY THE Z80 MONITOR IF AN INVALID COMMAND IS ENTERED.

COMMAND SET:	syntax
D(UMP)	D<START>,<END>
M(EMORY)	M<ADDRESS>
F(ILL)	F<START>,<END>,<BYTE>
C(OPY)	C<SOURCE_START>,<SOURCE_END>,<DEST_START>
V(ERIFY)	V<SOURCE_START>,<SOURCE_END>,<DEST_START>
T(EST)	T<MEMORY_START>,<MEMORY_END>,<DRAM BANK>
G(OTO)	G<ADDRESS>
I(NPUT)	I<PORT#>
O(UTPUT)	O<PORT#>,<DATA>
S(WITCH)	S (switch to 68000 CPU)
X(SELECT)	X<8-8" 5-5">,<0,1 -Side>,<1-S 2-D Density>
Y(DMA)	Y<DMA READ/WRITE ADDRESS> FE00 DEFAULT
Z(SECTOR)	Z<SIZE> 80 HEX DEFAULT
R(EAD)	R<UNIT>,<TRACK>,<SECTOR>
W(RITE)	W<UNIT>,<TRACK>,<SECTOR>
B(OOT)	B<DRAM BANK#> (boot up cpm or CYPHER-DOS)

The Z80 and 68000 monitors have various inter-processor slave commands that allow one CPU to give tasks or commands for the other CPU to execute. For example, the Z80 may command the 68000 to do a memory block move, or the 68000 may command the Z80 to do disk I/O. (See listings of the monitors and the CPM PLUS and CPM68K BIOS)

```

+++++
+      E P R O M   R E S I D E N T      +
+                                         +
+      A D M - 3 A ,   T E K T R O N I X   4 0 1 0      +
+                                         +
+      T E R M I N A L   E M U L A T O R S      +
+                                         +
+      P L U S   C Y P H E R   N A T I V E   C O M M A N D   S E T      +
+                                         +
+++++

```

A graphics terminal emulator written for the 68000 in C is included in the EPROM (monitor) set provided with the CYPHER. Characters may be transmitted to the driver through a jump table entry in both the Z80 monitor and the 68000 monitor (see source listings). The driver emulates an ADM-3A character terminal, a subset of the TEK4010 graphics terminal commands, as well as some special CYPHER native functions. The default configuration is an 80 character by 24 line format (480 x 216 pixels, no interlace) with approximately 10 screen pages simultaneously in memory. The following table describes the control characters and escape sequences used by the CYPHER graphics driver. Hexidecimal numbers are denoted with a trailing H.

----- A D M- 3 A Commands -----

ctrl-H	- backspace
ctrl-I	- tab
ctrl-J	- line feed (auto scroll up at bottom of page)
ctrl-K	- move cursor up (* AUTO SCROLL DOWN AT TOP OF PAGE *)
ctrl-L	- advance cursor right
ctrl-N	- carriage return
ctrl-X	- clear to end of line
ctrl-Z	- clear screen and home cursor
ctrl-~	- home cursor
esc = L C	- move cursor to line L-20H, col C-20H (absolute)
esc + L C	- move cursor by L-20H lines, C-20H cols (relative)

----- T E K 4 0 1 0 Commands -----

esc GS Yms Yls Xms Xls

- enter TEK4010 graphics mode and place cursor at pixel position X,Y where 0,0 corresponds to the bottom left of the screen. The four address bytes are determined as follows:

Yms = most significant 5 bits of Y + 20H
 Yls = least significant 5 bits of Y + 60H
 Xms = most significant 5 bits of X + 20H
 Xls = least significant 5 bits of X + 40H

Maximum values depend on the screen size as determined by the configuration command, esc R. Straight lines are drawn to subsequent positions entered as Yms Yls Xls only (no esc GS) while in graphics mode. Short addresses are supported (ie. may enter 1s bytes only) and the cursor is turned off. The

cursor can be repositioned without drawing a line by entering the esc GS (plus address) command again. (GS = 1DH)

Example:

esc GS 20h 60h 20h 40h	- move to X=0, Y=0, no line drawn, cursor off
30h 60h 50h	- draw line to X=16, Y=412
esc US	- exit graphics mode, cursor on

esc s	- draw with solid lines (default)
esc b	- draw with dashed lines
esc g	- draw with dotted lines
esc US	- exit graphics mode, turn cursor on (US = 1FH)
esc FF	- exit graphics mode, clear screen, turn cursor on (FF = 0CH)

C Y P H E R N A T I V E Commands

esc R Yms Yls Xms Xls

- set display size to Y + 1 pixels across (16 to 1024), X + 1 pixels down (1 to 1024) where X and Y are entered as described above for the esc GS command. The video clock speed (CONFIG register) and whether interlace is employed or not is determined automatically, but note that not all monitors will display all combinations. A size of 216 x 480 gives a 24 x 80 character display, 432 x 1008 gives a 48 x 168 display. Pixels across must be a multiple of 16. If used in character mode, X and Y should also be a multiple of 6 and 9 respectively for proper display.

esc S f	- set SLANT characters ON if f = "1", OFF if f = "0"
esc Z n	- set character write ZOOM factor to n-30H (0 to 15)
esc V f	- set video INVERT ON if f = "1", OFF if f = "0"
esc P n	- advance screen by n-30H pages (approx -10 to 10, depending on screen display size - NOTE FORWARDS and BACKWARDS capability)
esc T f	- draw during retrace only if f = "1", always if f = "0"
esc C f	- cursor ON if f = "1", OFF if f = "0". Character speed can be increased significantly if the cursor is turned off and drawing occurs any time.

esc @ Yms Yls Xms Xls - move cursor absolute to pixel location X,Y where the least significant 5 bits of Yms, Yls, etc, are used to generate a 10 bit address. 0,0 is the screen bottom left.

esc % Yms Yls Xms Xls - move cursor relative by X,Y pixels where the least significant 5 bits of Yms, Yls, etc, are used to generate a 10 bit two's complement value

 Note : The following software released in disk format has generally
 been superceded by the resident graphics package outlined above.

Terminal emulation functions on the Nec 7220 128k byte graphics provided on
 disk as GCYPHER.COM (source is also provided):

COMMAND VALUE	FUNCTION
------------------	----------

1	^A	normal video
2	^B	inverse video
3	^C	home cursor
4	^D	char. cursor goto x,y char. pos. (OFFSET 20 hex for x,y 2 bytes)
5	^E	clear to end of line
8	^H	backspace
9	^I	tab
10	^J	linefeed
12	^L	clear screen
13	^M	carriage return
16	^P	set line pattern for all future line drawing
19	^S	draw line from pixel cursor position to x,y absolute (4 bytes)
20	^T	draw line from pixel cursor position to +x,+y relative (4 bytes)
21	^U	move pixel cursor to position x,y absolute (4 bytes)
22	^V	move pixel cursor to position +x,+y relative (4 bytes)

GCYPHER.COM file provided will place the Z80 nec7220 terminal emulation
 programme in static memory under CP/M-80 as the list device. By entering cntl-
 p you will see your input echoed on your video monitor.

EPROMS/CPM INTERFACING:

=====

Two Eproms are provided providing 16k of software. One contains the odd bytes,
 the other the even bytes in the 16 bit data path. The even byte eprom is
 installed at F23, and the odd byte eprom at E23.

We have provided a sample BIOS for the CYPHER on diskette.

16K Eprom memory allocation:

Z80 monitor	- FE0000-FE0FFF (4 K)
68K monitor	- FE1000-FE1FFF (4 K)

The remaining 8K may later be assigned to eprom 7220 terminal emulation. The
 terminal emulation programme is presently loaded from disk under CPM-80.

GENERATING A NEW VERSION OF EPROMS:

ASSEMBLE Z80 MONITOR:

Enter the following to assemble and link the Z80 monitor:
(using Microsoft assembler)

```
M80 =Z80MON                      (assemble Z80 monitor)
L80 /P:O,Z80MON,Z80MON/N/X/E     (link it)
N                                 (reply N to move into memory)
```

ASSEMBLE 68k MONITOR:

If you are using the CPM users group 68k assembler (will not assemble graphics terminal emulator code):

```
A68K 68KMON,=68KMON.A68
MBASIC SREC                      (convert S records to assembler file using
                                  BASIC program)
ASM file name given in SREC.BAS
You now have a .hex file for DDT
```

If you are using the CPM-68K assembler (this is the preferred method and allows the graphics terminal emulator to be included):

```
AS68 -L 68KMON.S                 (assemble 68k monitor into 68kmon.o)
AS68 -L GRAPHI.S                 (assemble graphics terminal emulator code
                                  generated with the CPM-68K C compiler)
AS68 -L FONTTAB.S               (assemble fonttable output generated by the
                                  GENFONT.C program under CPM-68K)
L068 -TFE1000 -B3FE00 68KMON.O GRAPHI.O FONTTAB.O
                                  (create absolute file for loc. fe1000)
Output is placed in c.out file.
```

You now need to create the even and odd eproms using cpm-80 DDT.

*Use
splitter.com
instead*

```
DDT
I68KMON.HEX
R6000
IC.0UT                          (or I68KMON.HEX)
R
M11c,111c,7000                 (Enter this for CPM-68K assembler use. The
124 11FF                       the offset ignores the header created by
                                CPM-68K.)
```

We are now ready to split the code residing in 6000-7FFF HEX to 2000-2FFF hex for even eprom and 4000-4FFF hex for odd eprom:

```
SPLIT16K                       (We have now split what was in 6000-7fff
                                into two regions; 2000-2FFF for even eprom
                                and 4000-4FFF for odd eprom.)
```

CPM INSTALLATION NOTES:

=====

Boot from CYPHER-DOS else you will need to have access to a friend's CP/M-80 system. After you have obtained your licensed CP/M-80 you may either recompile the Bios provided or patch our compilation for a 60k system.

CPM-80:

1. Use sysgen to create a 60k cpm-80 version.
2. Using DDT move Z80 image of boot in eprom to boot location. Or move in Z80boot.900 for a length of 100hex bytes to 900 hex.
3. Move Z80 bios to bios overlay area. Or read in Z80bios.1f8 to 1f80 hex.
4. Save sysgen along with image of cpm for later use.
5. Execute the sysgen containing the new cpm2.2.
6. Bypass first "get cpm prompt".
7. Give command to create CYPHER 60k cpm-80 on new diskette.

CPM-68k:

Read your licensed CP/M-68 manual first.

1. Under cpm80, use the basic programme SREC.BAS to convert SR400.sys S-records to an 8080 assembler file. This will allow you to load the S-records and create a 8080 hex file. If you do not have a basic interpreter, then you need to use the 68k monitor S-record load command.
2. The SREC.BAS output file must be assembled using the 8080 cpm80 assembler creating a 8080 hex file. You are prompted by SREC.BAS for the name of the output ASM file, and the origin. (Examine the output file.)
3. Use CPM-80 DDT to load the hex file created in step 2.
4. Save the SR400 hex file as a com file of 32 K bytes. (Uses less than that.) You now have a file that CPM-80 can load.
5. Using cpm-80 DDT load in SR400.COM. This file is an absolute file
6. Move 100-8000 hex to 400.
7. Read in 68kBIOS.SYS to 6000hex. This will overlay and place the bios.
8. Patch in the start of bios in SR400.sys as indicated in your CPM68k manual. This may be at location 4f98hex. The ccp entry point has been set to 4f9C hex in the 68k-bios. Check if this is different for your version of CPM68k. Refer to your CPM-68k manual for bios addresses.
9. Assemble at 100 hex:
OUT FF ; this will switch to 68k cpu monitor
JMP F02A ; this will place the Z80 in slave mode
10. Save 128 cpm68k.com
11. Under CPM-80 in any bank enter the command "cpm68k". This will load in the CPM-68k image.
The boot at 100-3ff will branch to the 68k-monitor. Under the 68k monitor, issue the copy command to copy the modified SR400 to 400.
Enter the Go command to 400 (G400).

Subsequent versions of CPM68K can be generated by assembling and linking the CPM68k bios (BIOS68.S) as per the instructions in the CPM-68k manual.

The following only applies to the disk based Z80 graphics code.

Video using nec7220 section:

Enter GCYPHER executing GCYPHER.COM. This file is configured at 1C5 HEX with the following defaults (Order a NEC 7220 manual from your Dealer if required):

1C5	06 - op. mode
1C6	12 - Active display words per line - 2 , bit 0 = 0
1C7	09 - bit 0-4 Horiz sync width - 1 (1 to 32 chars.) bit 5-7 Vert Sync width low order bits (1 to 32 horz lines)
1C8	0E - bit 0-1 Vert sync width high order bits (1 to 32 horz lines) bit 2-7 Horz. front porch width - 1 (1 to 64 chars.)
1C9	07 - bit 0-5 Horz. Back porch - 1
1CA	01 - bit 0-5 Vert. Front Porch width
1CB	F0 - bit 0-7 Active display lines per Video field, low bits
1CC	05 - bit 0-1 Active display lines per Video field, high bits bit 2-7 Vertical Back porch Width

The NEC7220 requires 8 bytes of initialization. these are located at 1C5 hex in the .COM file. This will make the graphics section the LST: device for cpm (enter cntrl p to activate the list device).

Examples:

For a 20 MZ monitor with min. 10 ms horiz. blank., 5 ms vertical, 800 ms blanking, and 4-32 ms horiz. sync. you can enter 0F,32,64,18,04,02,2C,31 at location 1C7 in the GCYPHER.COM file.

A file on the Distribution diskette called GDCTEST.COM will test the 128k GDC memory. Use this file to test the viability of the graphics memory.

SECTION 14 CONSTRUCTION TIPS

=====

Choose a well lighted work space with enough room to place all parts and tools. Make sure you that you have enough lighting and comfort. Familiarize yourself with the manual and all steps before beginning. I may be wise to go through the parts list and steps, making sure that you understand all the steps.

If you have never constructed a circuit board, try to obtain help from someone experienced.

Tools required:

1. A 20 WATTS or less soldering iron, preferably temperature controlled.
2. A soldering iron holder will prevent you from burning yourself.
3. Wire cutter to cut excess wire.
4. A pair of pliers to bend leads.
5. A wet sponge to clean and wipe the soldering iron of excess solder or resin.
6. A magnifying glass to examine any errors in soldering.
7. ROSIN CORE SOLDER (NOT ACID CORE!) with 63/37 tin lead ratio.
8. If you can afford it, a desoldering pump can help you correct your mistakes.
9. EYE PROTECTION GLASSES MUST BE WORN WHEN SOLDERING OR CUTTING WIRE.

Make yourself comfortable, set your chair at a proper height. Clean off your work area before and during your work. Pieces of metal may get lodged under a IC socket and ruin your day.

A solder splash can cause a short. A hot iron can damage PC traces, diodes, transistors and pads. Hold your iron enough to solder but not burn the pad or place excess solder.

You should wipe your soldering iron with the wet sponge or cloth every 10-20 connections.

NEVER solder the CYPHER while its plugged in.

NEVER use cleaning solutions (it may desolve the board!).

NEVER use a worn out soldering tip.

NEVER pull on a joint to test the connection (use a OHMmeter).

NEVER flip excess solder from the tip of your iron (use the wet sponge or cloth).

NEVER put solder on your iron and then place it to a joint (this will result in a cold solder joint).

Heat both the component wire and the pad with the tip of the iron until it looks wet or liquid. Then touch the solder to the junction between the iron, the pad, and the wire. When the solder melts and flows onto the connection, quickly remove the iron's tip. Allow the joint to cool without disturbance. A good joint will be smooth and shiny. A bad one will be dull and "blob like". Beware of flying pieces when cutting or soldering, wear eye protection.

Limited Warranty:

Motel Computers Ltd. ("MOTEL") warrants this CYPHER Z80/68000 single board computer against defects in materials and workmanship to the original purchaser for a period of Ninety (90) days from date of purchase, subject to the following conditions.

What is covered by this warranty

During the ninety (90) day warranty period MOTEL will, at its option, repair or this MOTEL product or repair or replace with new or used parts any parts or components, manufactured by MOTEL, which prove to be defective, provided the product is returned to an Authorized MOTEL Dealer as set below.

How to Obtain Warranty Service

In order to obtain warranty service, first obtain a return authorization number by contacting the Authorized MOTEL Dealer from whom you purchased the product. Then attach to the product:

1. Your name, address and telephone number.
2. the return authorization number,
3. a description of the problem, and
4. proof of the date of retail purchase.

Ship or otherwise return the product, transportation and insurance costs prepaid, to the Authorized MOTEL Dealer. If you are unable to receive warranty repair from the Authorized MOTEL Dealer from whom you purchased the product, you should contact Motel Computers Ltd.

WHAT IS NOT COVERED BY THIS WARRANTY

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THIS WARRANTY AND THE STATUTE OF LIMITATIONS SHALL RUN CONCURRENTLY WITH ANY ACCEPTANCE PERIOD. THIS WARRANTY IS NOT TRANSFERABLE. NO SUIT, LITIGATION, OR ACTION SHALL BE BROUGHT BASED ON THE ALLEGED BREACH OF THIS WARRANTY OR IMPLIED WARRANTIES MORE THAN ONE YEAR AFTER THE DATE OF PURCHASE.

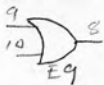
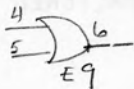
THIS WARRANTY IS VOID FOR BARE BOARDS OR KITS EXCEPT FOR COMPLETENESS OF PARTS, QUALITY, OR FOR UNALTERED OR UNASSEMBLED PARTS.

Possible Spare LS08

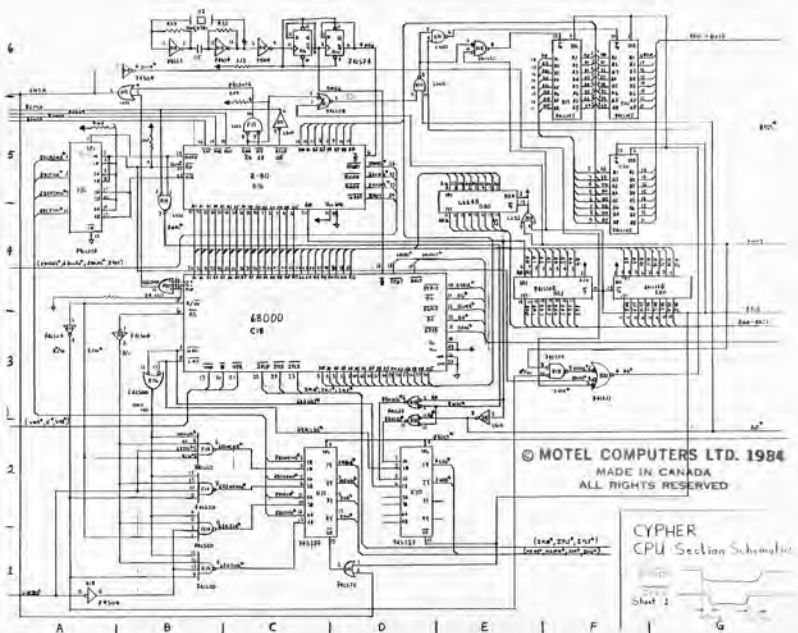
H30 pins 11, 12, 13

F3 pins 1, 2, 3 ; 4, 5, 6

Spare LS32



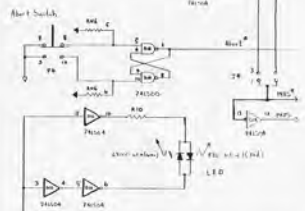
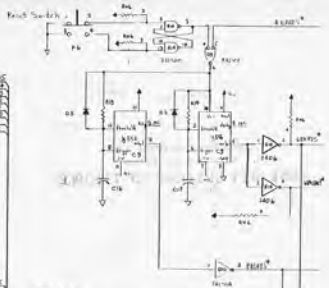
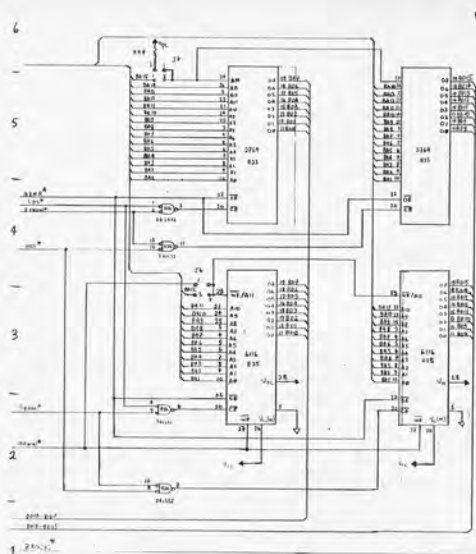
D14 - LS02b, c, d I think.



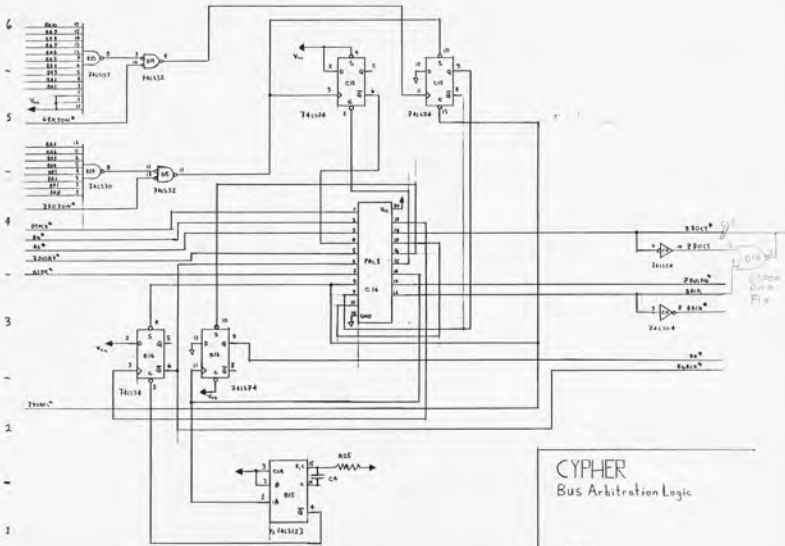
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CYPHER
CPU Section Schematic

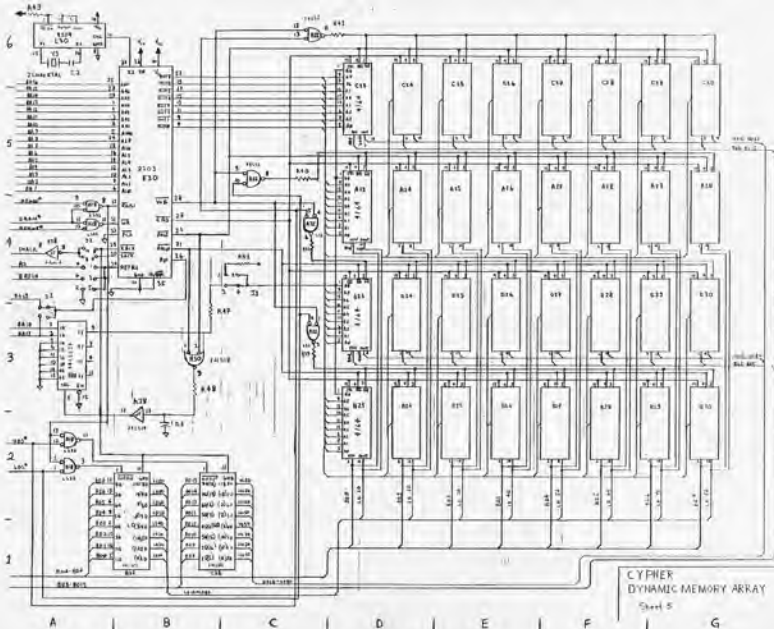
Sheet 1

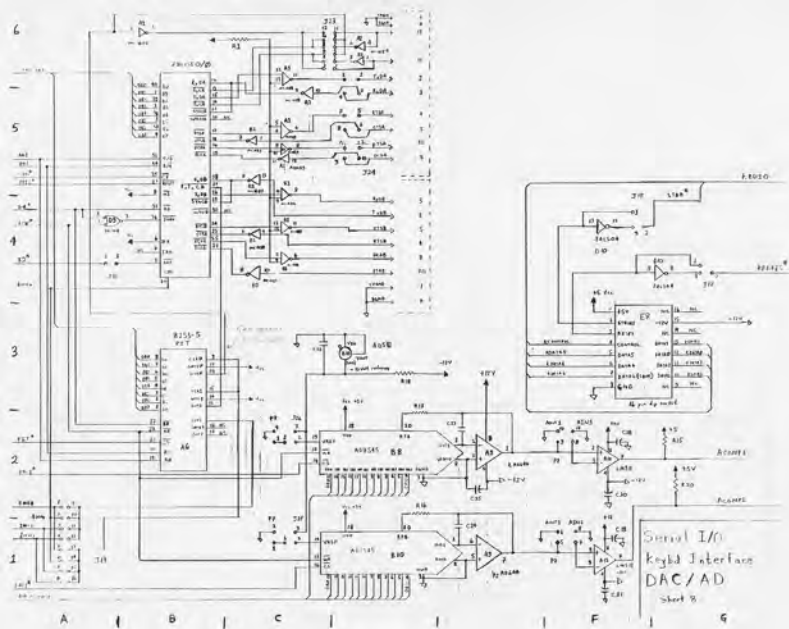


EPROM/5RAM
 Reset; Abort
 Processor Action L=1
 Sheet 3



CYPER Bus Arbitration Logic





6

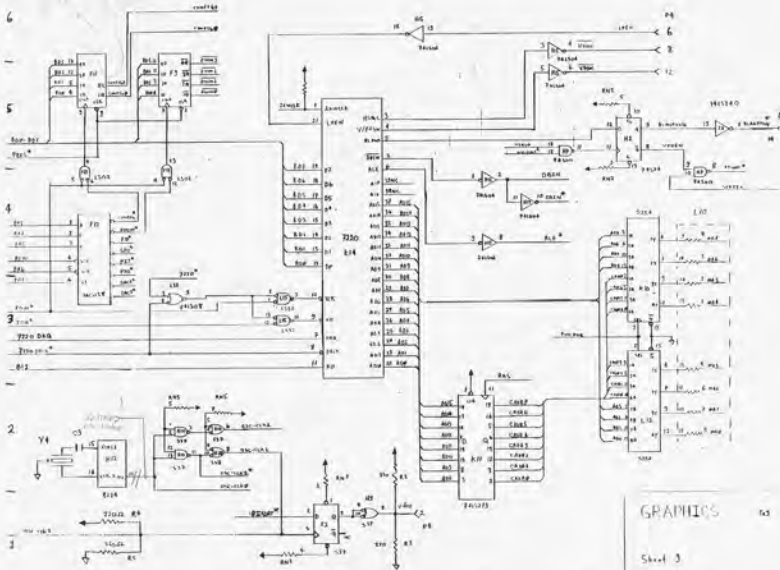
5

4

3

2

1

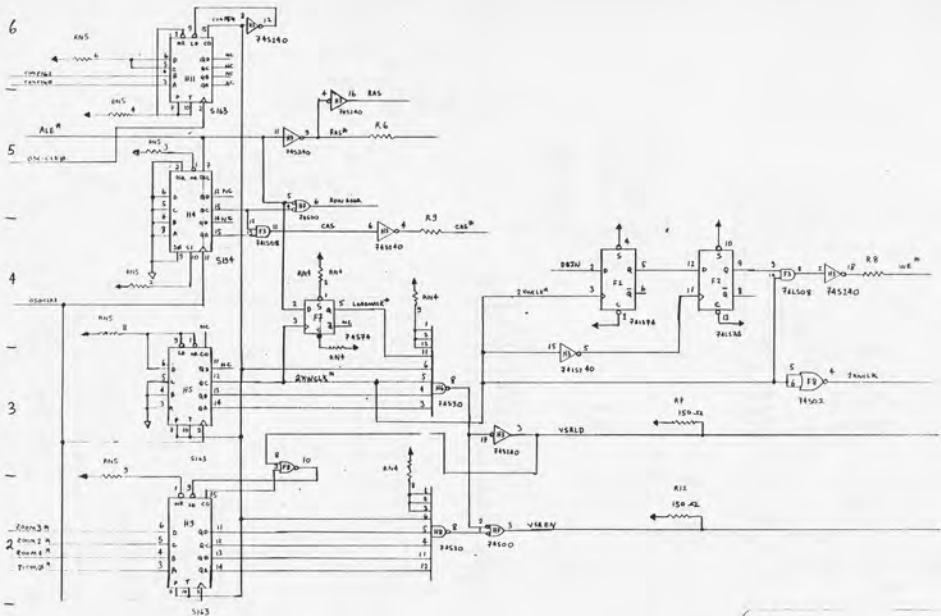


GRAPHICS

65

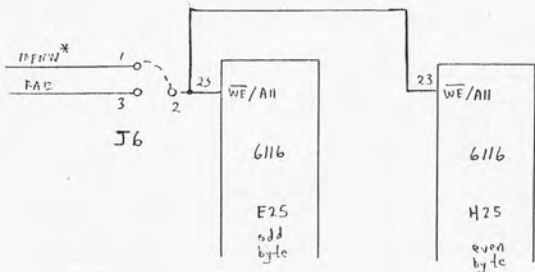
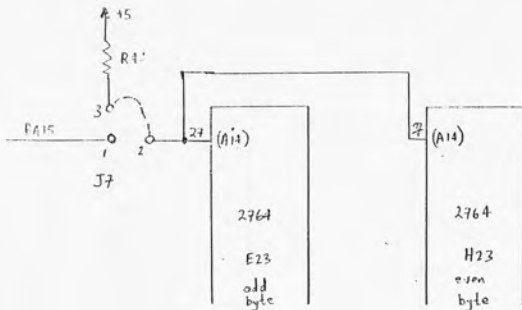
Sheet 3

A | B | C | D | E | F | G



Graphics 62

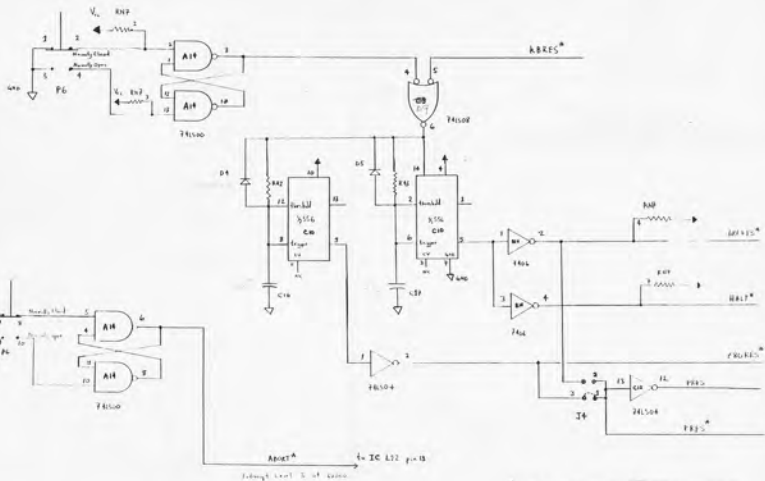
Sheet 10



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EPROM / SRAM
Address Jumping
CIPHER V5.1.1
MOTEL COMPUTERS
1984

a



RESET, ABORT CIRCUIT

64-100000-10.1.1

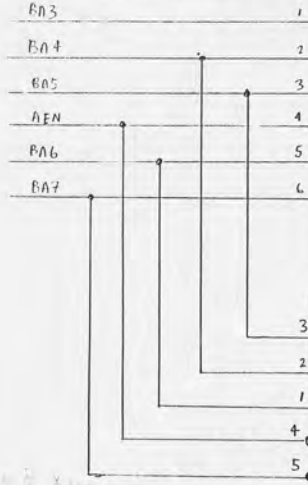
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Model Computer 1324

B

Note: Address lines start from BA1 (FEF AB not used)



8011-FFH

74LS138	Y3	12	CONFIG*	99 (FEF 299)
A	Y2	13	ZOOM*	91 (FEF 291)
B	Y7	7	FD*	D9-BF (FEF 2B9, B, C, F)
C	Y6	9	SIO*	E8-BF (FEF 2B8)
G2A	Y5	10	PIT*	19, 4B, 4D, 4F FEF 8 (A9, AB, AD, AF)
G2B	Y4	11	PIO*	A1, A3, A5, 17 (FEF BA1, 3, 5, 7)
G1	Y0	15	DAC1*	FEF 820
E12	Y1	14	DAC2*	FEF 288

74LS138	Y0	15	PSELECT *	01 (FEF 801)
C	Y1	14	7220 *	41, 43 (FEF 841, FEF 843)
B	Y2	13	DMA *	1X (FEF 81X)
A	Y3	12	NC 5X	To expansion board
G2A	Y4	11	NC 3X	
G2B	Y5	10	NC 6X	
G1	Y6	9	RTC *	(31, 33) (FEF 831, FEF 833)
K13	Y7	7	NC 7X	

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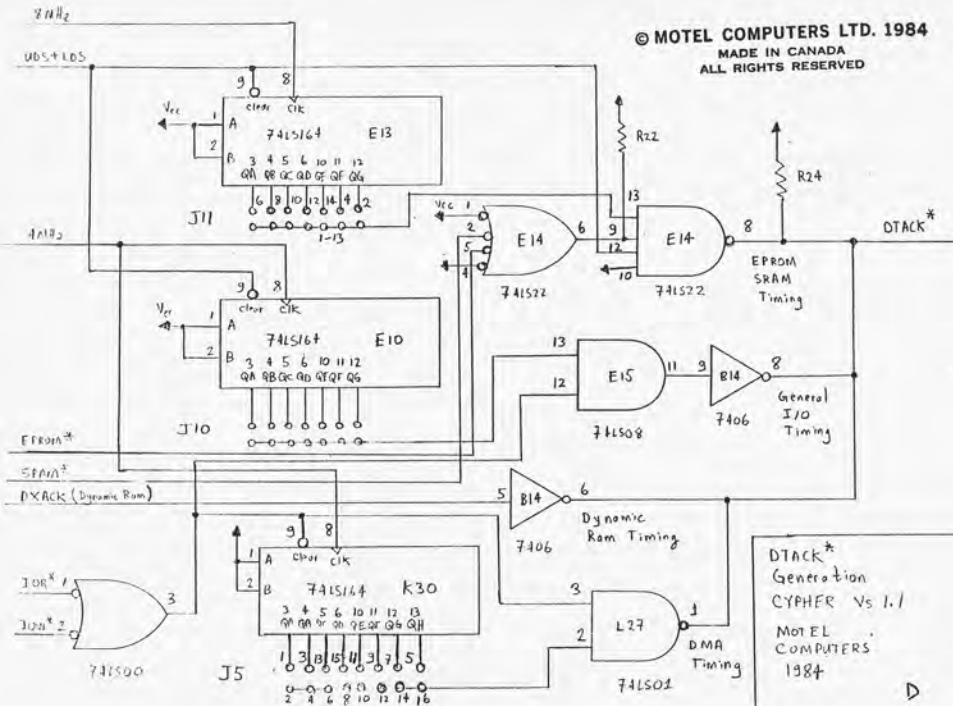
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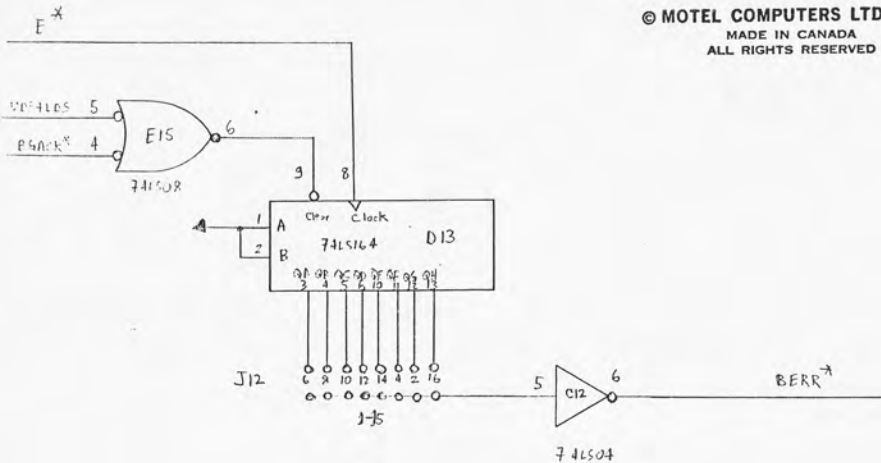
Peripheral Decoding
CYPHER V3 1.1

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Peripheral Decoding
CYBER V3 1.1
MOTEL COMPUTERS
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BERR* Generation

CYPHER Vs 1.1

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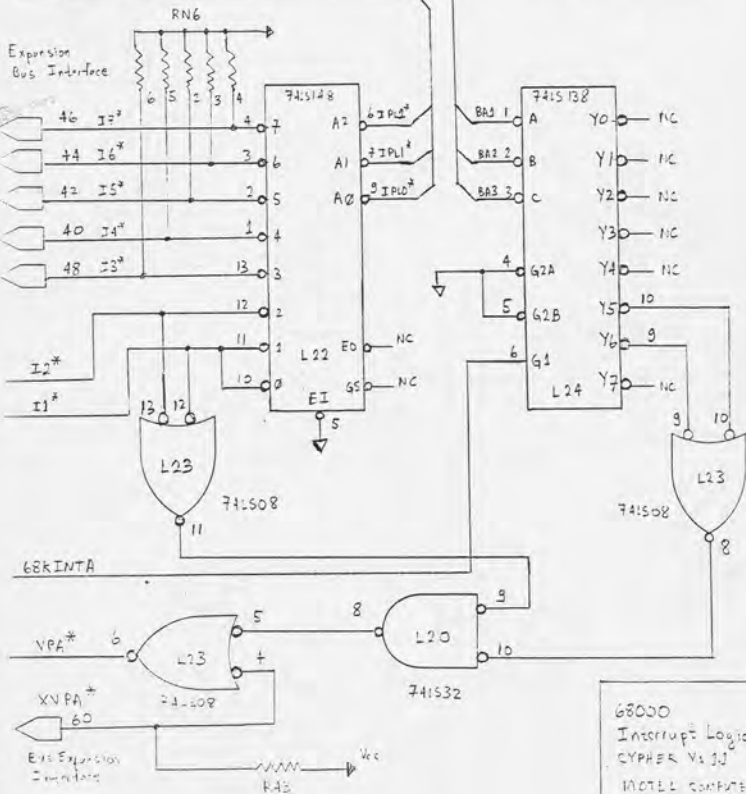


BA1, BA2, BA3

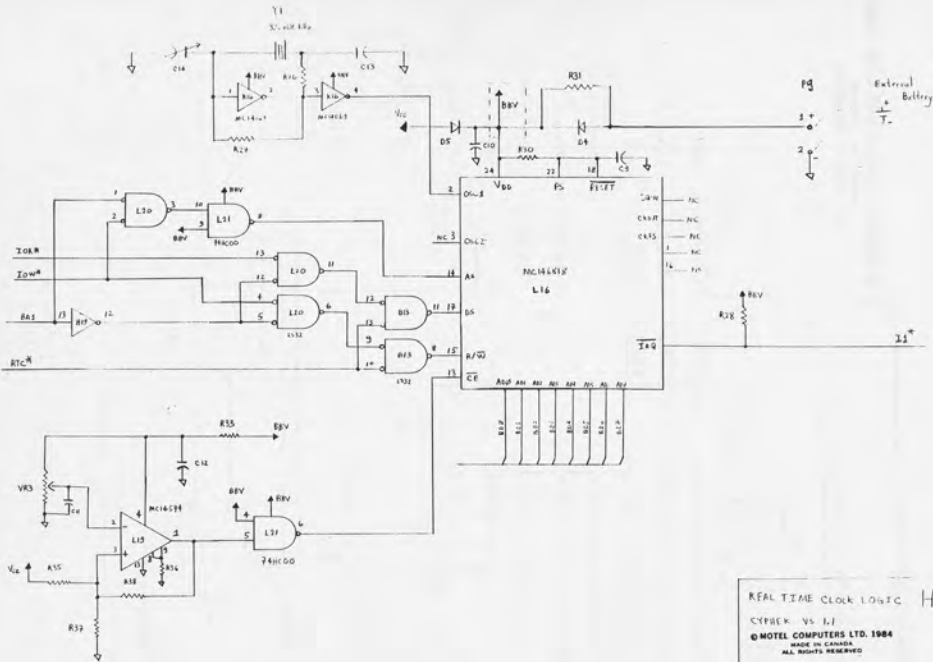
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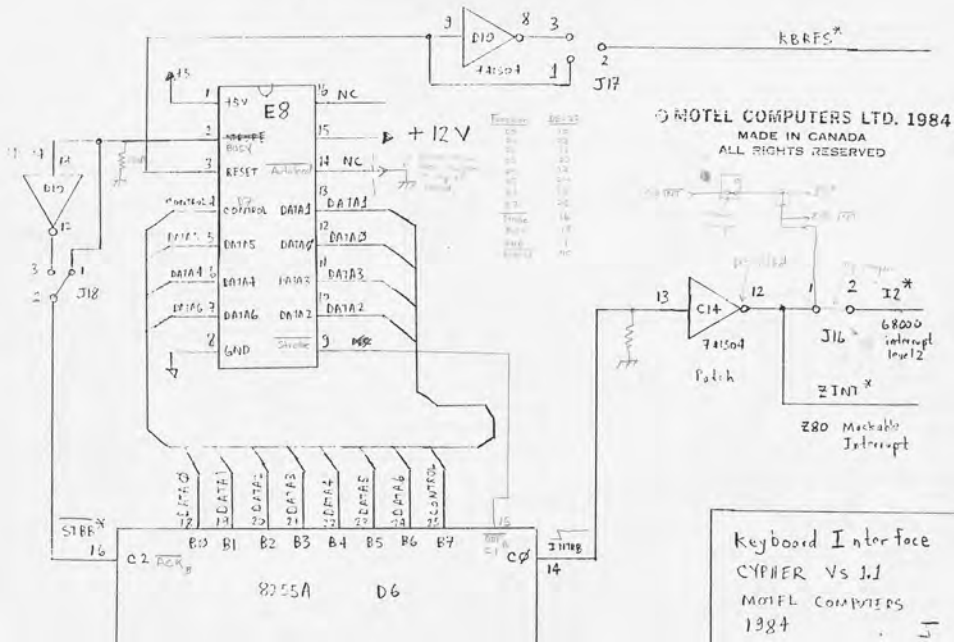
$IPLO^*$, $IPL1^*$, $IPL2^*$ (68000 input)



68000
Interrupt Logic
CYRUS V. 1.1
MOTEL COMPUTERS
1984

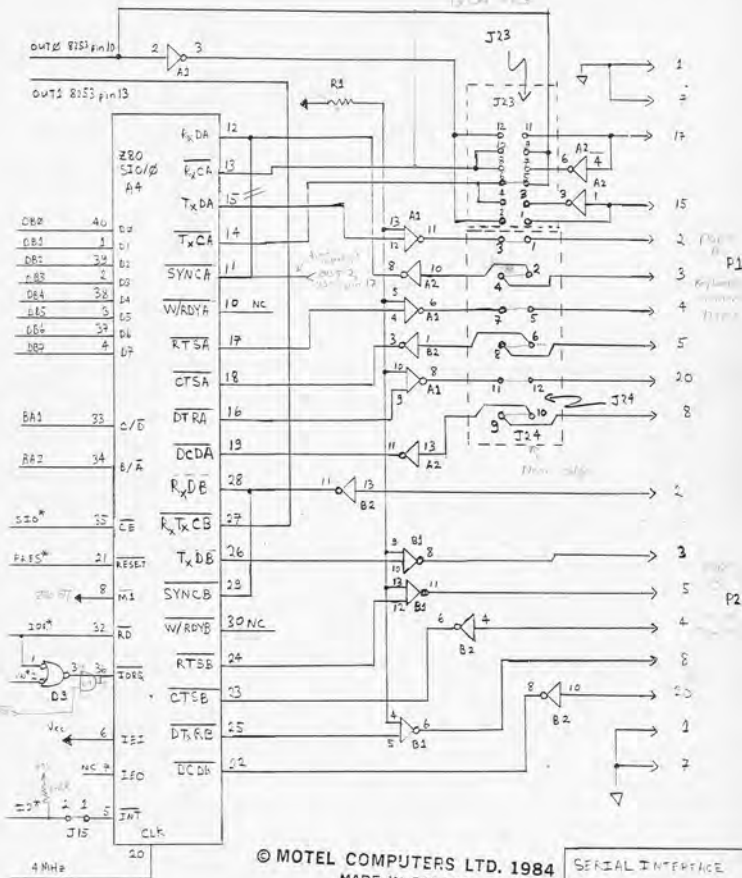


Keyboard and Mouse to printer Port



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Keyboard Interface
CYPHER Vs 1.1
MODEL COMPUTERS
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SERIAL INTERFACE
CYPRESS VC 4.1
MOTEL COMPUTERS
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BD0 - BD7

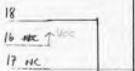
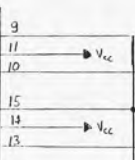
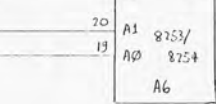
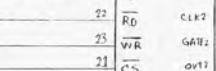
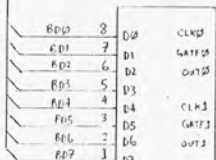
JOR*

JOW*

JIT*

PA2

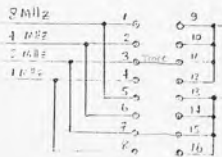
PA1



TxCA Bow Clock For SIO channel A
+ RxCA

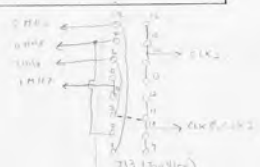
TxRxCB Bow Clock For SIO channel B

→ Slave of SIO-pin 11 (Timer Interrupt)



J13

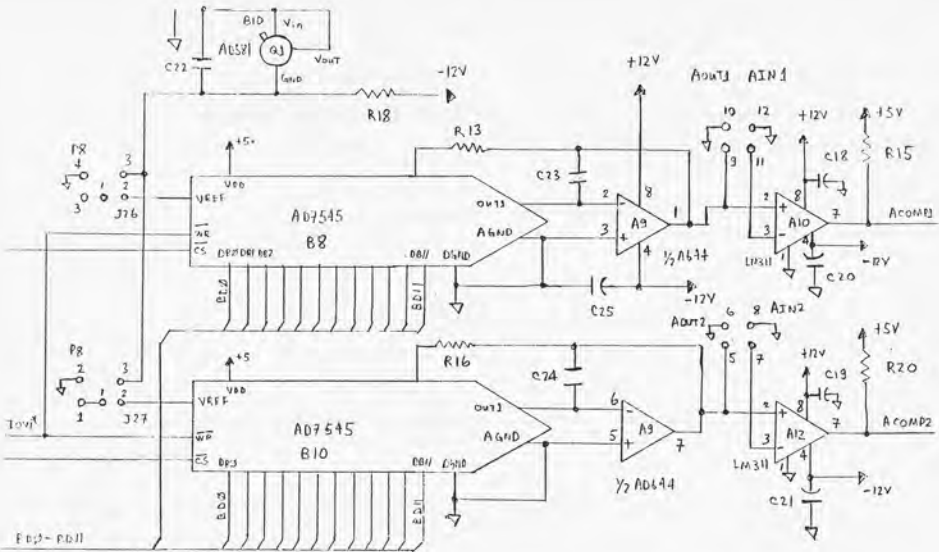
ALL Wires



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Programmable
Interval Timer
CYBER VS 1.1
MOTEL COMPUTERS
1984

L



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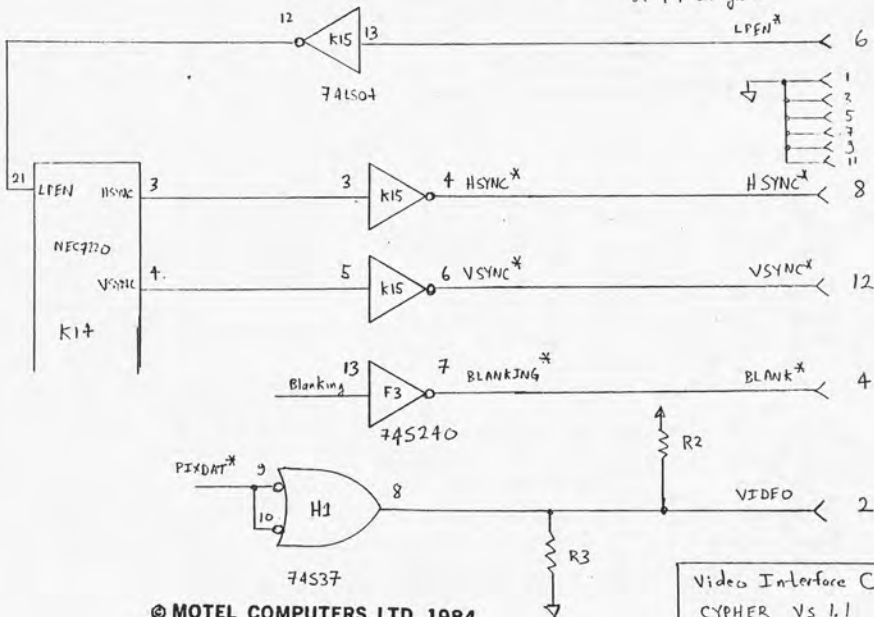
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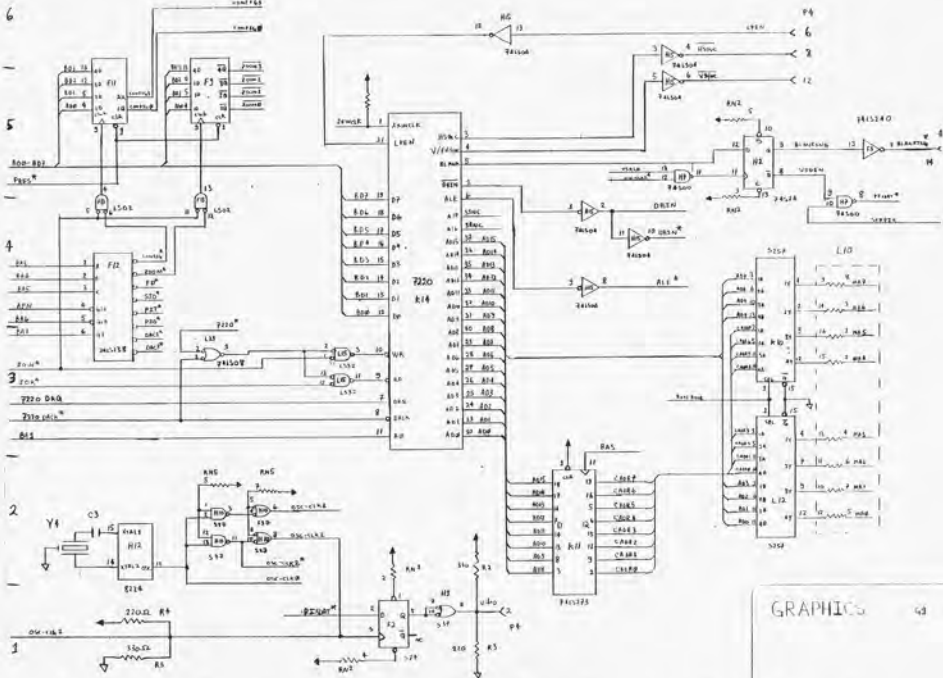
NOTE: All 7-pin pins on P8 are ground

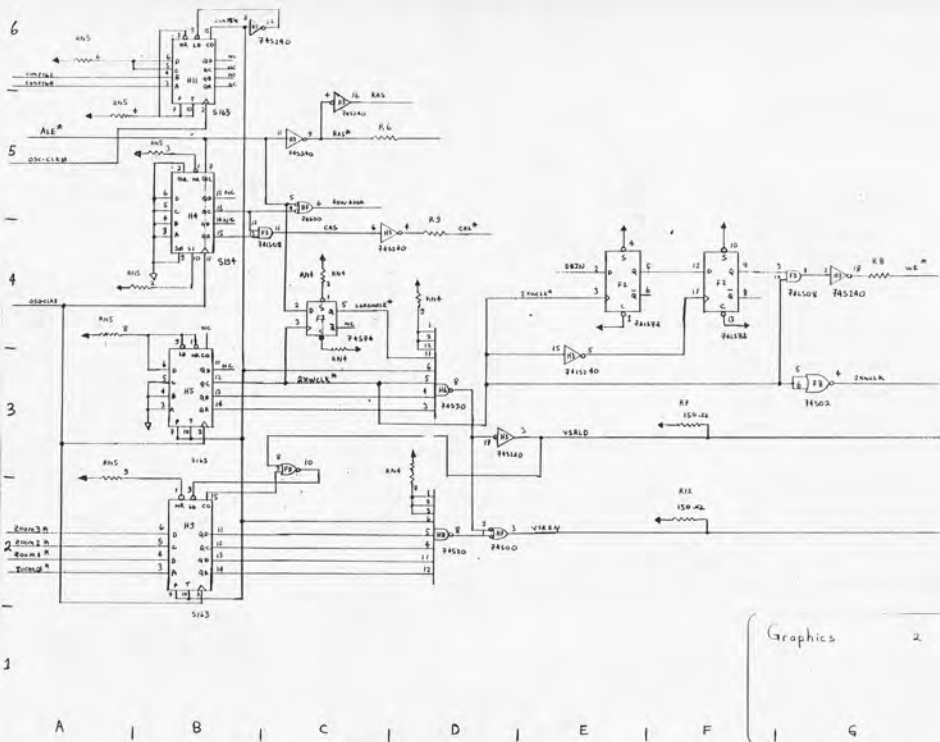
DAC / AD Interface
CYPRER V3.1.1
MOTEL COMPUTERS
1984

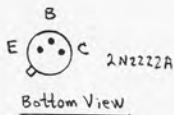
NOTE: All odd pins
of P4 are ground

P4

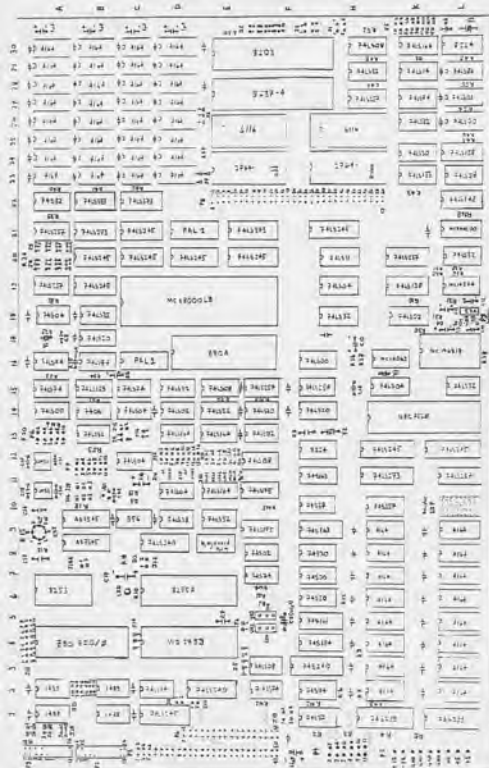








COMPOSITE
VIDEO CIRCUIT
(Add On)



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